

Advanced Soybean Agricultural Dataset (2025)

Business Research Management
Project Report

Mitzi Lanado

A. Description of the Dataset and Problem

The Advanced Soybean Agricultural Dataset (2025) is a high-resolution agronomic dataset curated for the purpose of understanding and enhancing soybean crop productivity. Developed through a collaborative research initiative at the College of Agriculture, University of Tikrit, this dataset integrates environmental, physiological, and agronomic variables to facilitate a data-driven approach to modern farming practices.

Dataset Structure and Variables:

The dataset initially contained over 55,000 rows collected across various field trials but was refined to 108 high-quality, non-redundant entries after comprehensive data cleaning and preprocessing. The key variables are categorized into the following groups:

- **Environmental and Soil Variables:**
 - *Rainfall, Temperature, and Soil Quality Metrics*: These determine growing conditions and influence physiological responses in plants.
- **Plant Physiological Variables:**
 - *Plant Height (PH)*: A general indicator of plant vigor and developmental progress.
 - *Leaf Area Index (LAI)*: Reflects the photosynthetic surface area, which directly correlates with growth rate.
 - *Relative Water Content in Leaves (RWCL) and Chlorophyll A (663)/B (649)*: Indicators of plant stress and photosynthetic efficiency.
 - *Sugars (Su) and Protein Percentage (PPE)*: Markers of nutritional content and metabolic activity.
- **Yield and Output Variables:**
 - *Biological Weight (BW)*: Total above-ground biomass of the plant.
 - *Number of Pods (NP), Number of Seeds per Pod (NSP), Weight of 300 Seeds (W3S)*: Key yield components.
 - *Seed Yield per Unit Area (SYUA)*: Final productivity metric used as the dependent variable in regression analysis.
 - *Protein Content (PCO)*: Indicates market quality of the harvested product.

Research Problem and Objective:

Soybean is a globally significant crop, and optimizing its yield is critical for both food security and economic sustainability. The primary goal of this project is to identify which biological and physiological traits most significantly affect soybean yield, with a

focused investigation into the relationship between Biological Weight (BW) and Seed Yield per Unit Area (SYUA).

The underlying hypothesis is that heavier, healthier plants (with higher BW) are more capable of efficiently converting their biomass into seed production, making BW a strong predictor of SYUA. If this is confirmed statistically, it provides farmers and agronomists a measurable target for crop improvement strategies.

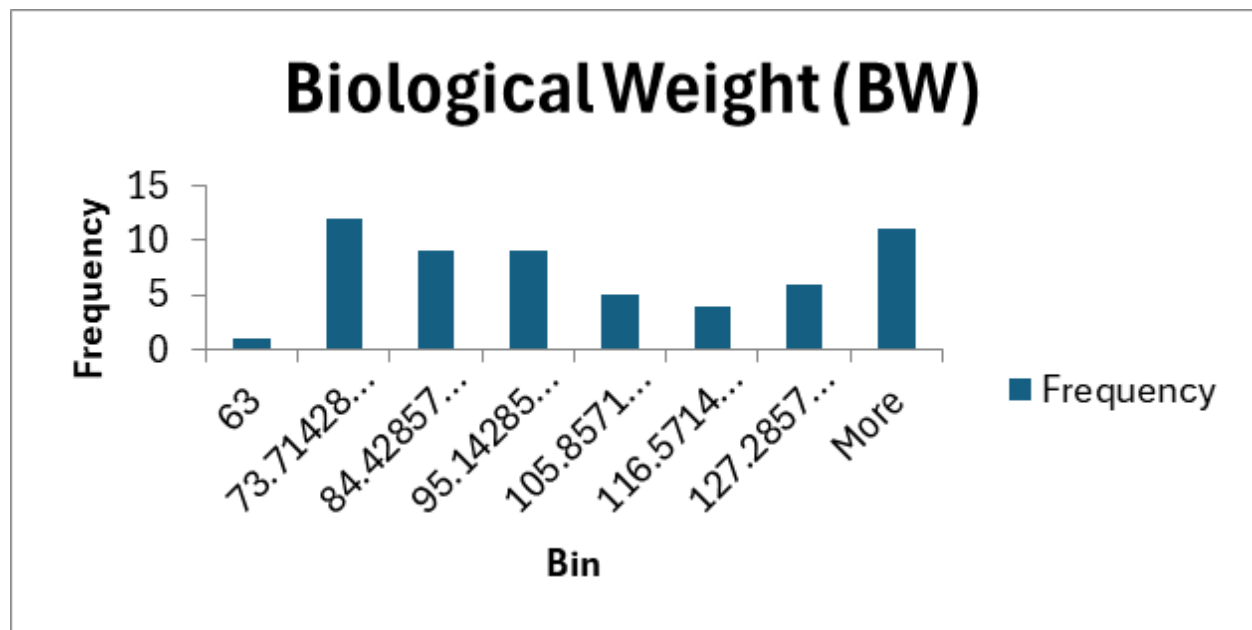
This problem aligns with broader questions in precision agriculture:

- Which traits can be measured early to predict final yield?
- How can these traits be enhanced through farming practices or breeding?
- What environmental or physiological indicators should be monitored to guide decision-making?

By combining traditional agronomic insight with statistical regression and hypothesis testing, this study aims to establish a reliable, data-supported basis for improving soybean crop yield at scale.

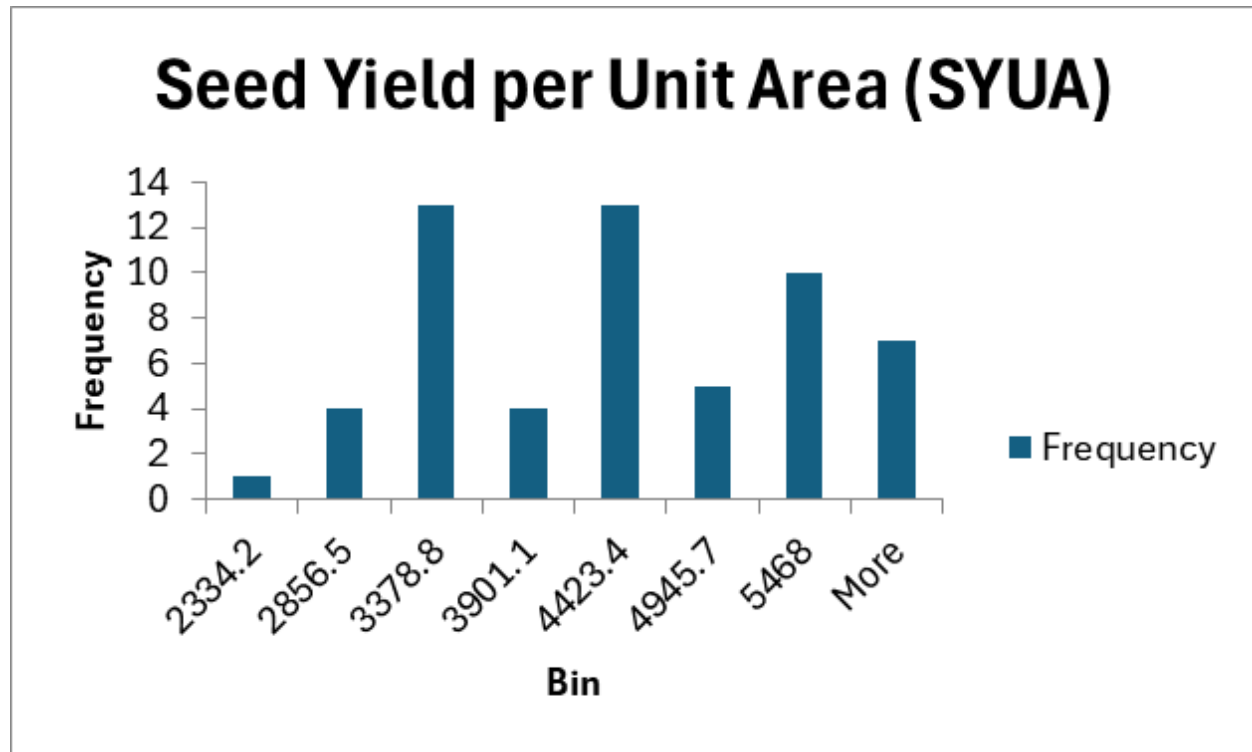
B. Visualizations and Analysis

B1. Histograms



The Biological Weight (BW), which is the independent variable of this report, shows a positive skew to the right. This shows that most soybeans have BWs between

73.71 to 95.14 units. Based on our dataset, only 1 has a BW that is lower than most. Moreover, approximately 11 soybeans have BWs heavier than the ranges shown above.

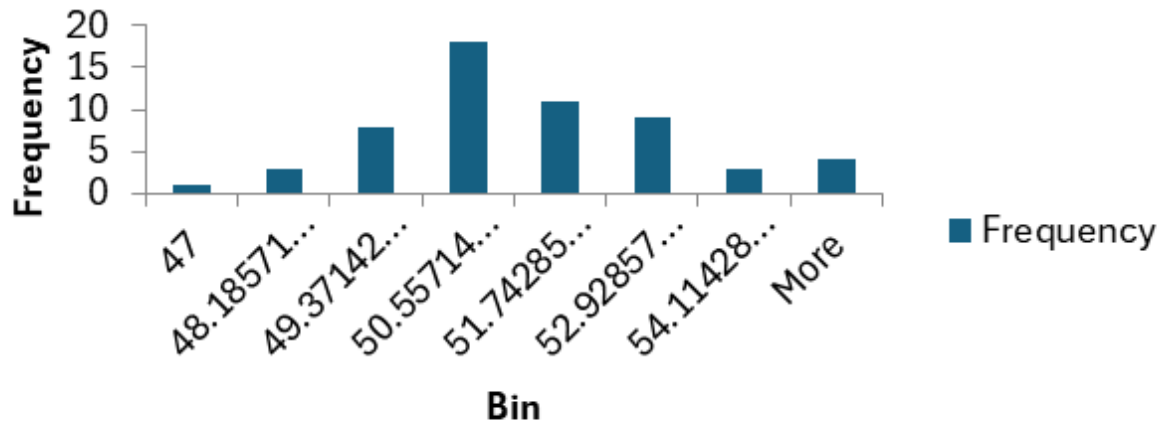


The Seed Yield Per Unit Area (SYUA), which is the dependent variable of this report, shows a hint of bimodal distribution with more variability. The most frequent SYUA values are along 3378.8 and 4423.4. The rest of the histogram shows fewer observations, meaning that most samples have an average SYUA.

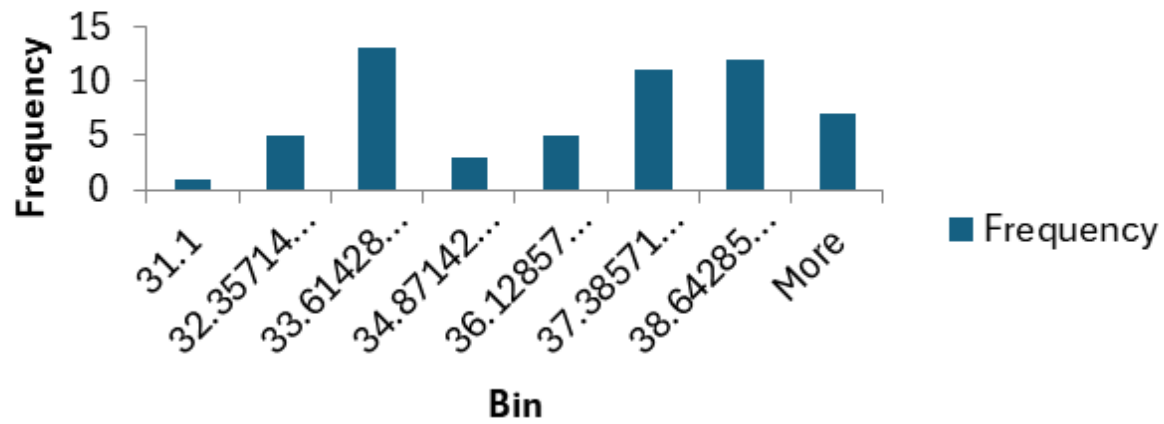
Symmetrical Distributions

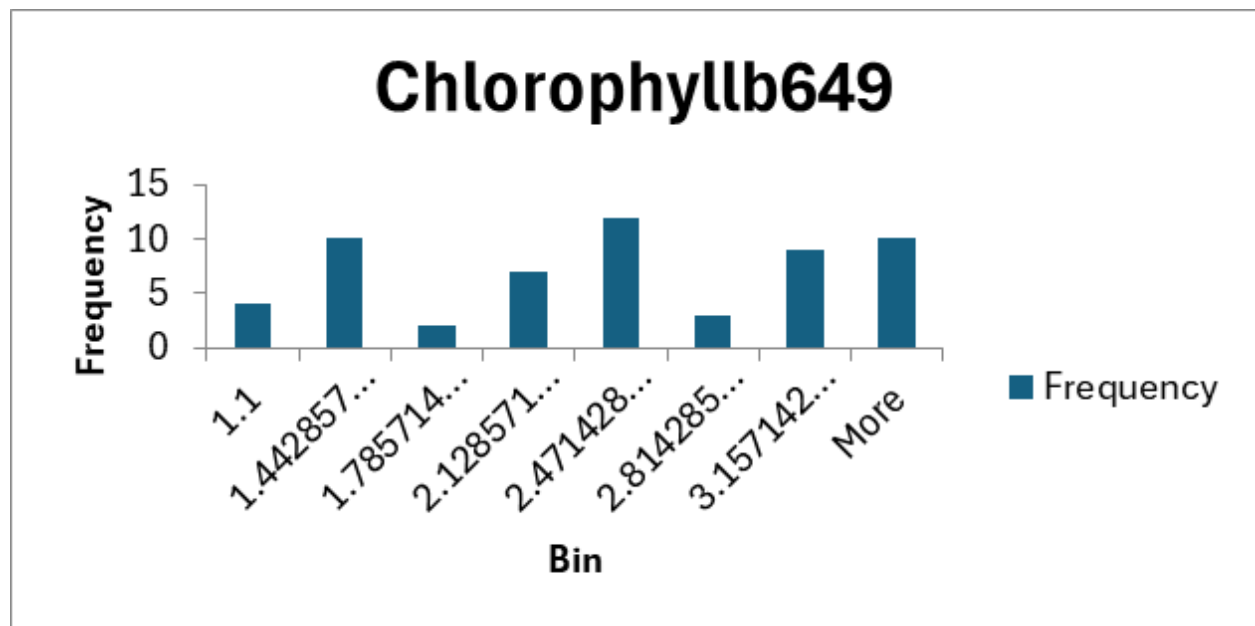
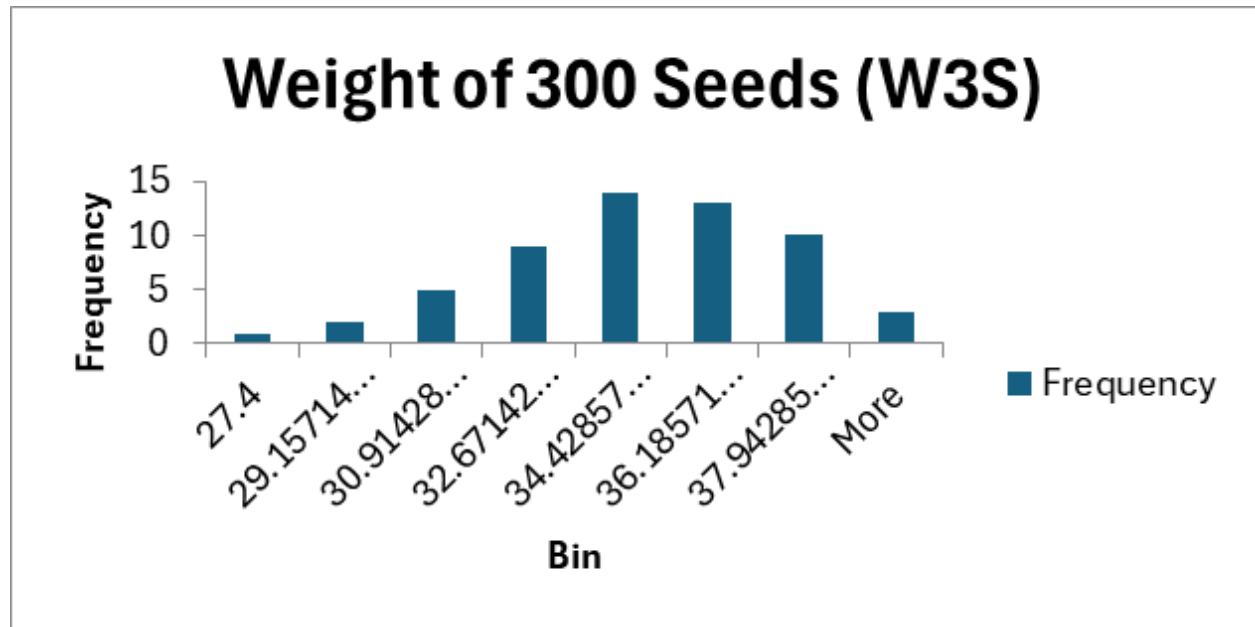
The Plant Height (PH), Protein Percentage (PPE), Chlorophyllb649, and Weight of 300 Seeds (W3S) have symmetrical distributions. PH and W3S show a normal distribution because of their bell shape, which means that most of the data is situated around their respective means. The PPE is symmetric, however, not normally distributed, with peaks at points 33.61 and 38.64. Chlorophyllb649 has a mild uniform distribution with frequencies alternating from low to high.

Plant Height (PH)



Protein Percentage (PPE)

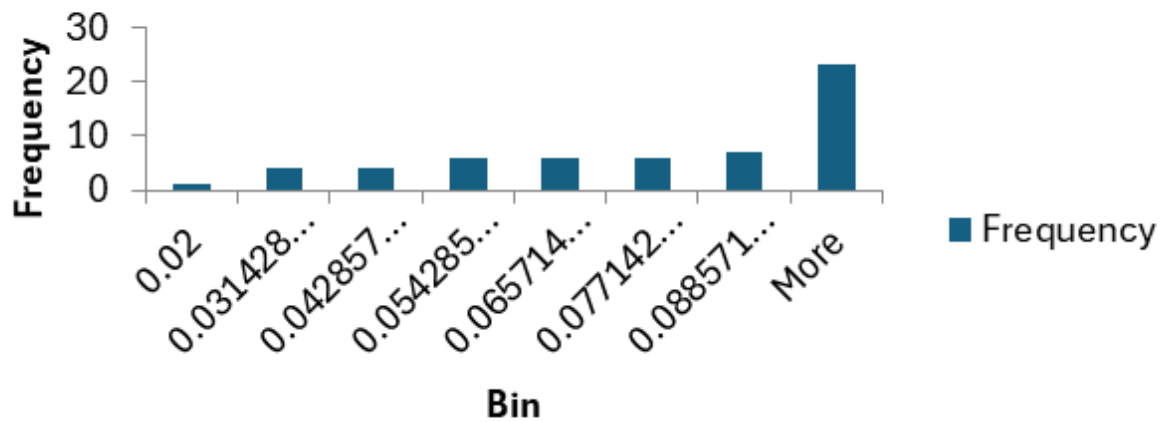




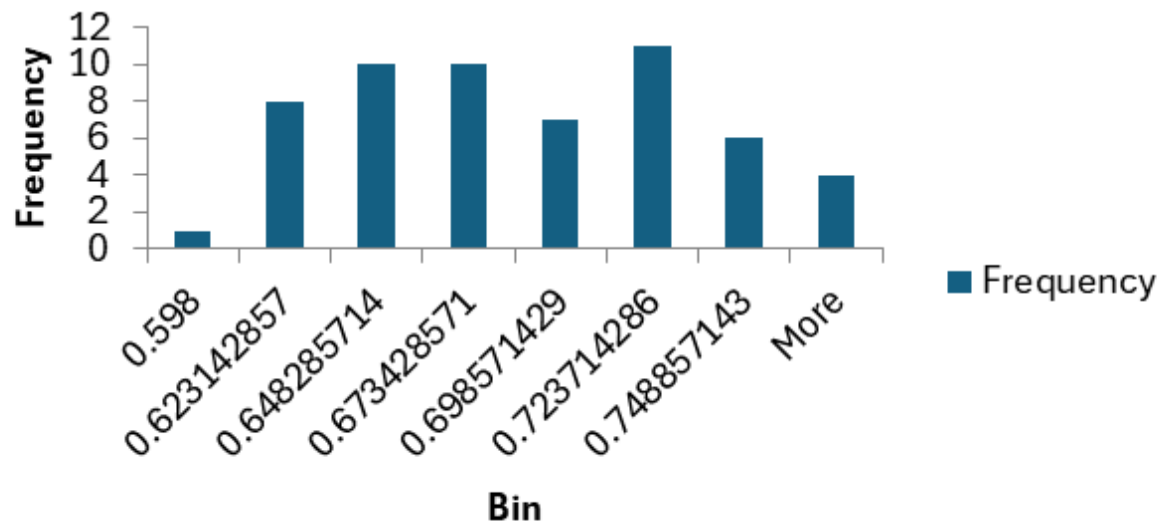
Right-Skewed

The Leaf Area Index (LAI) and ChlorophyllA663 have a heavy skew to the right, which means that most of their values are low with a few outliers. Relative Water Content in Leaves (RWCL), Number of Pods (NP), and Protein Content (PCO) have mild to moderate skews. RWCL has values clustered into the middle ranges, while PCO is spread across the bins and has higher frequencies in its middle ranges.

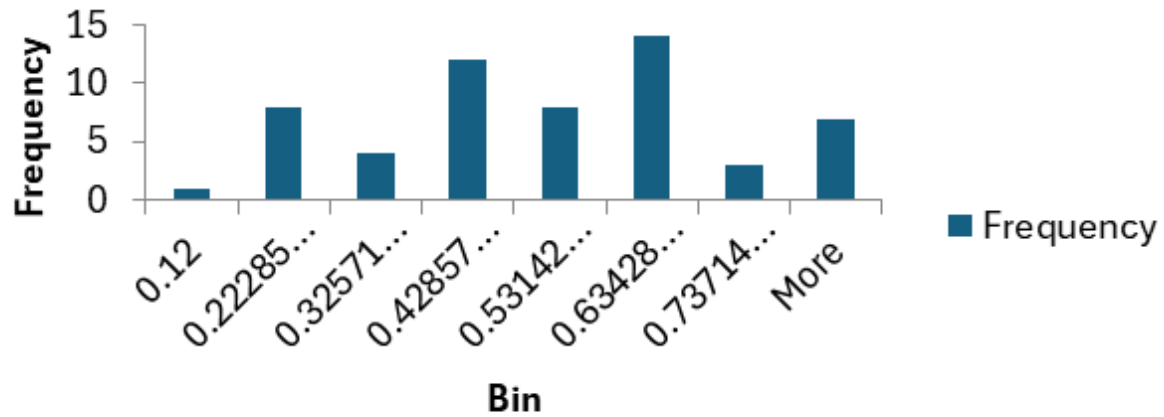
Leaf Area Index (LAI)



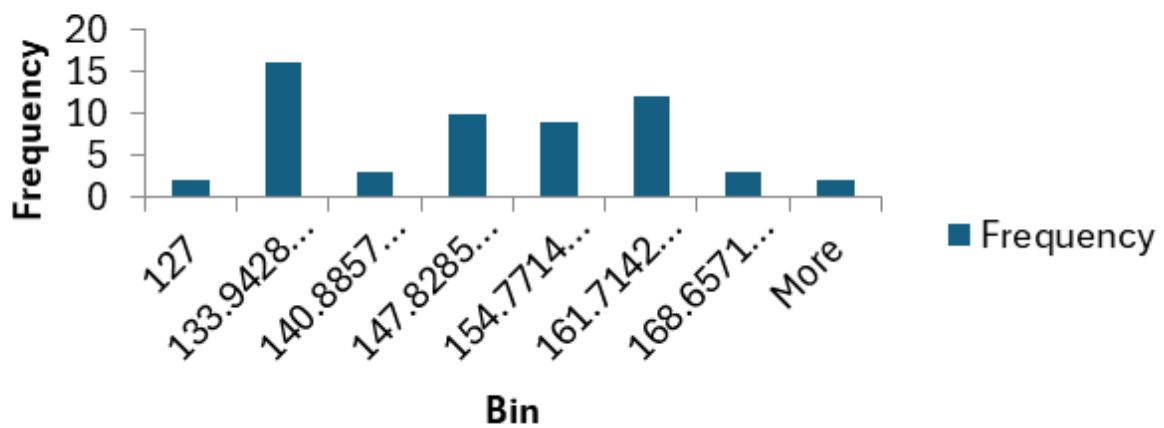
Relative Water Content in Leaves (RWCL)

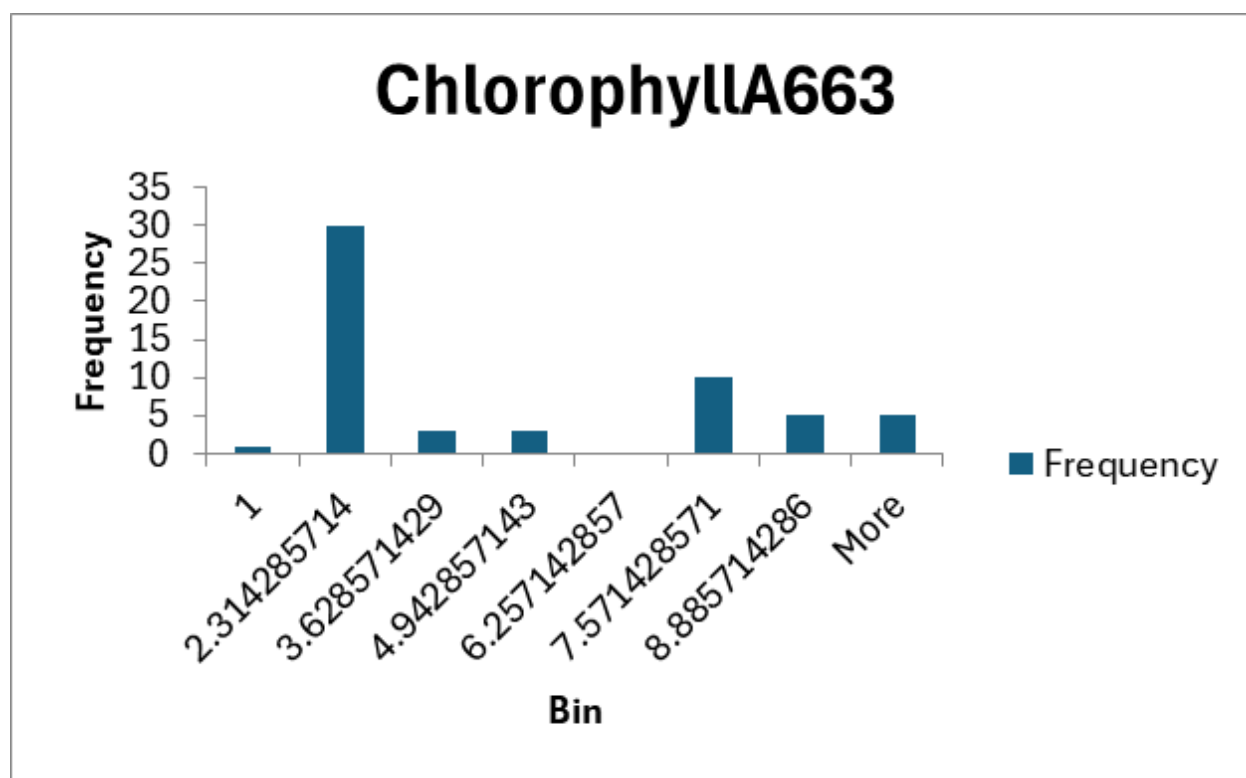


Protein Content (PCO)



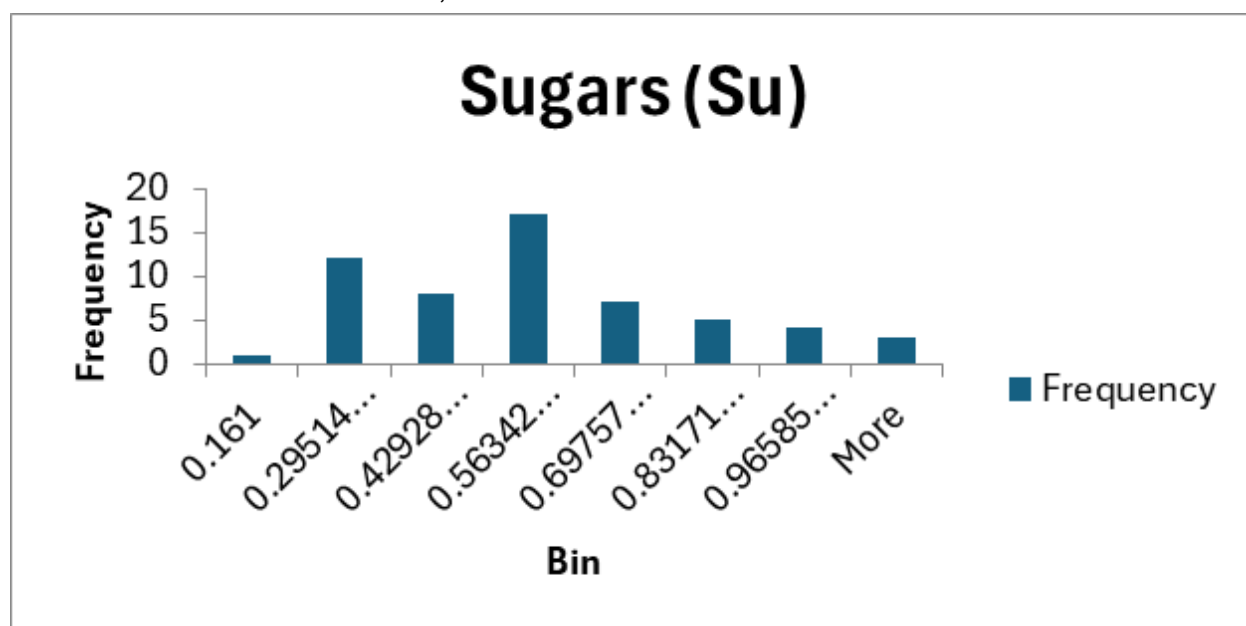
Number of Pods (NP)





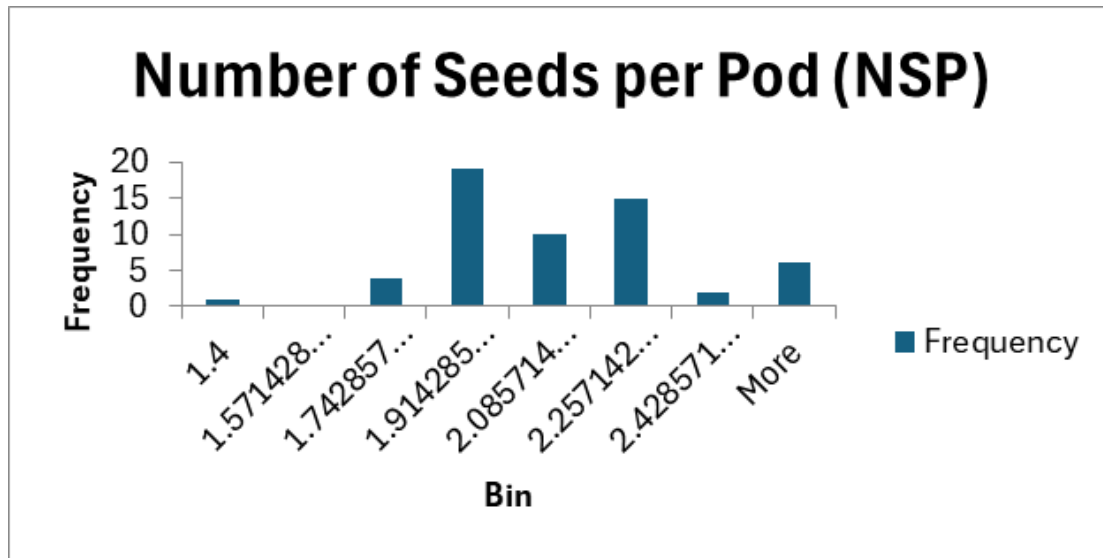
Left-Skewed

Sugars (Su) has a negatively left skew which means that most sugar values cluster towards the lower ends, which are 0.295 and 0.563.

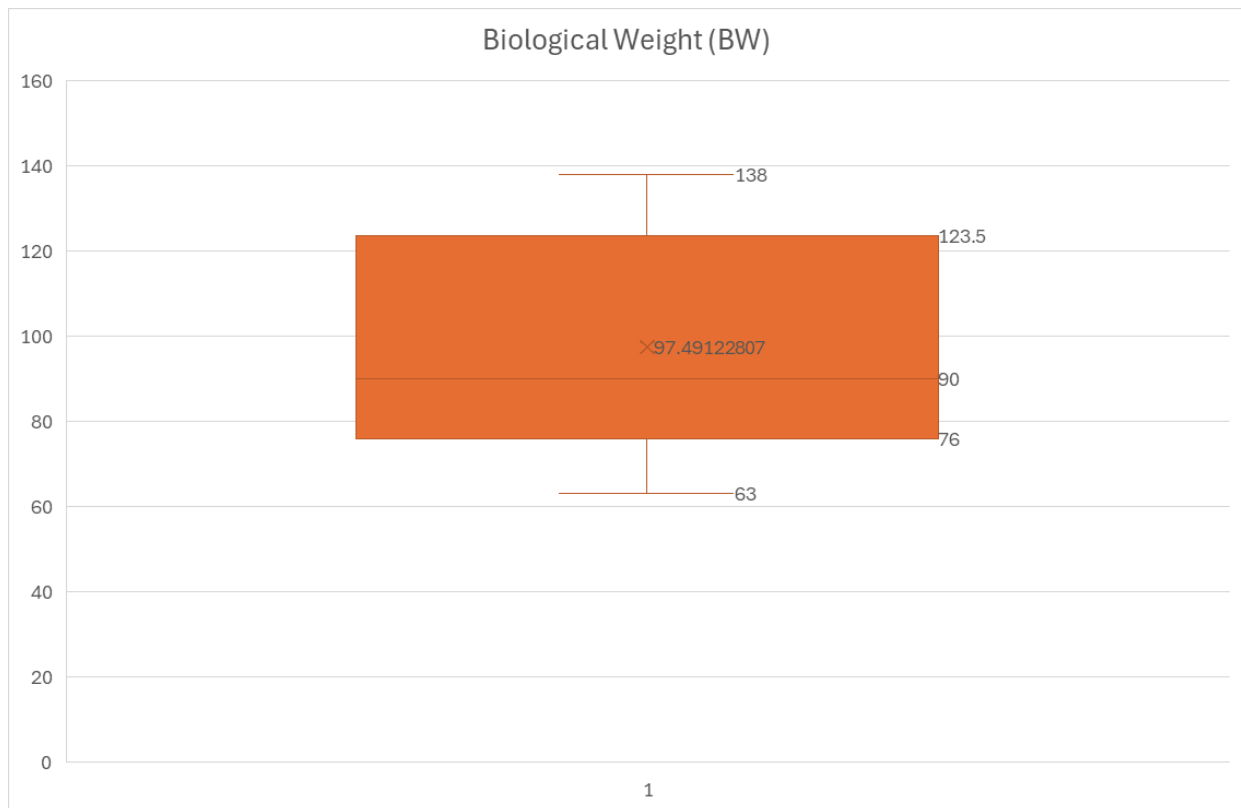


Bimodal

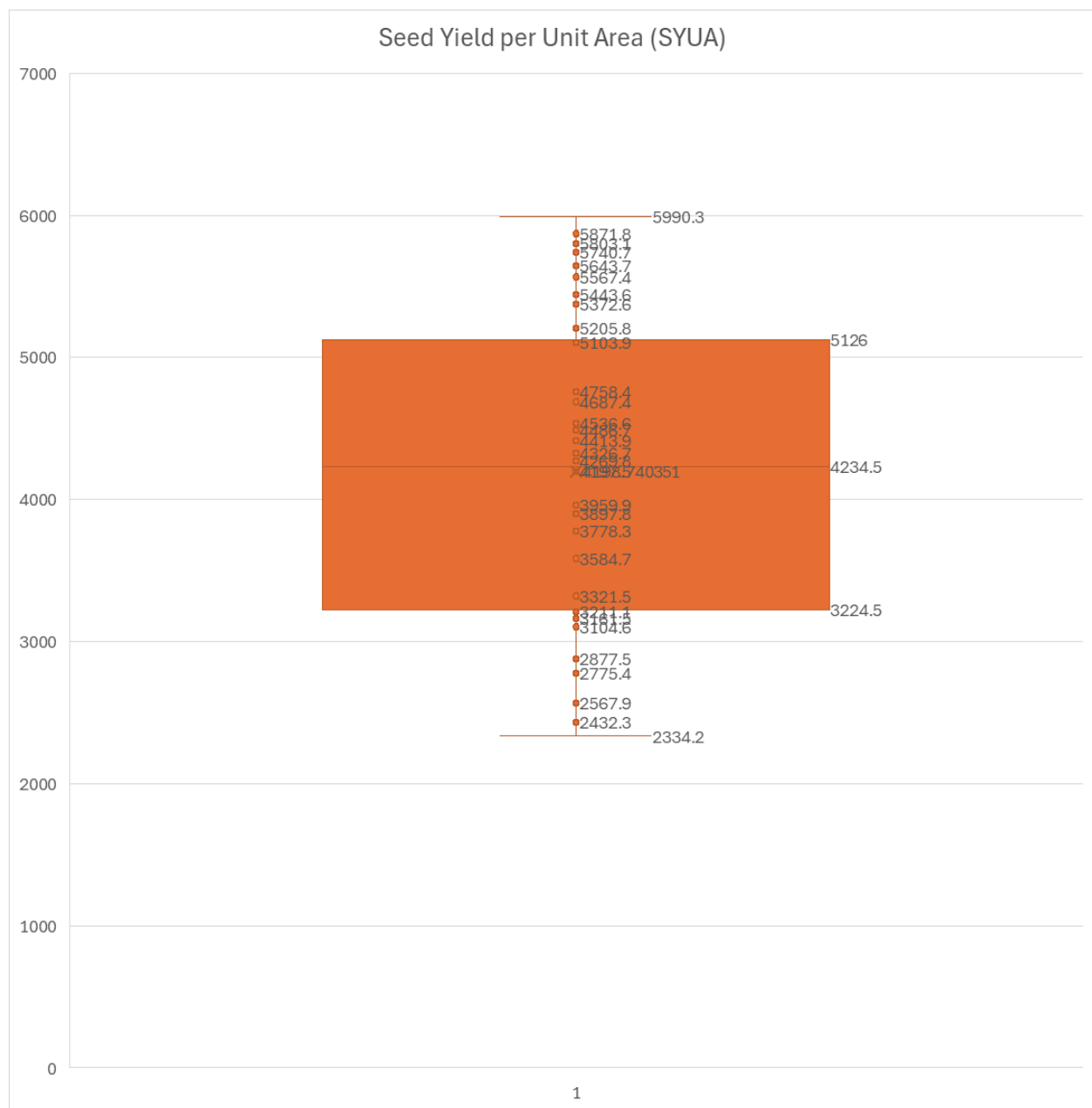
The Number of Seeds per Pod (NSP) has 2 peaks, 1.91 and 2.26, which shows that NSP's values drop outside of these peaks.

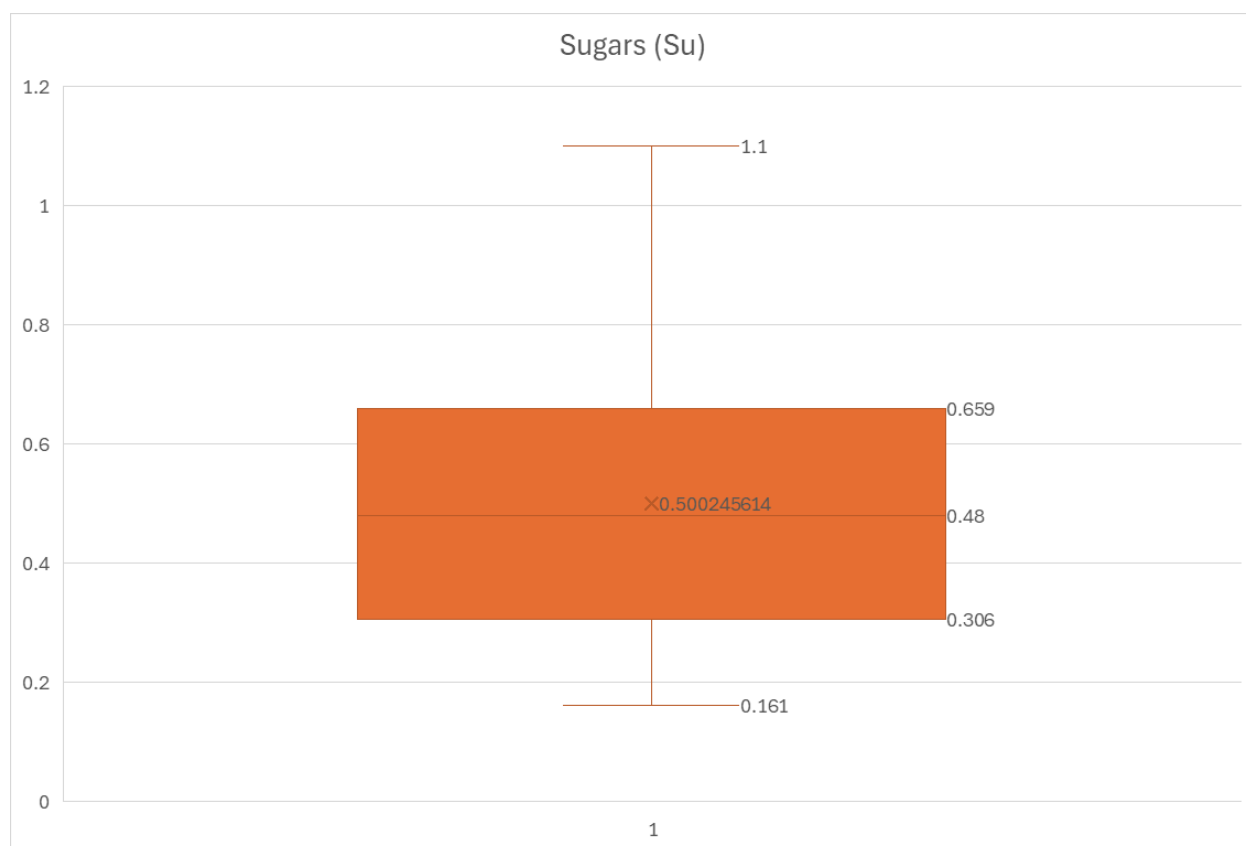
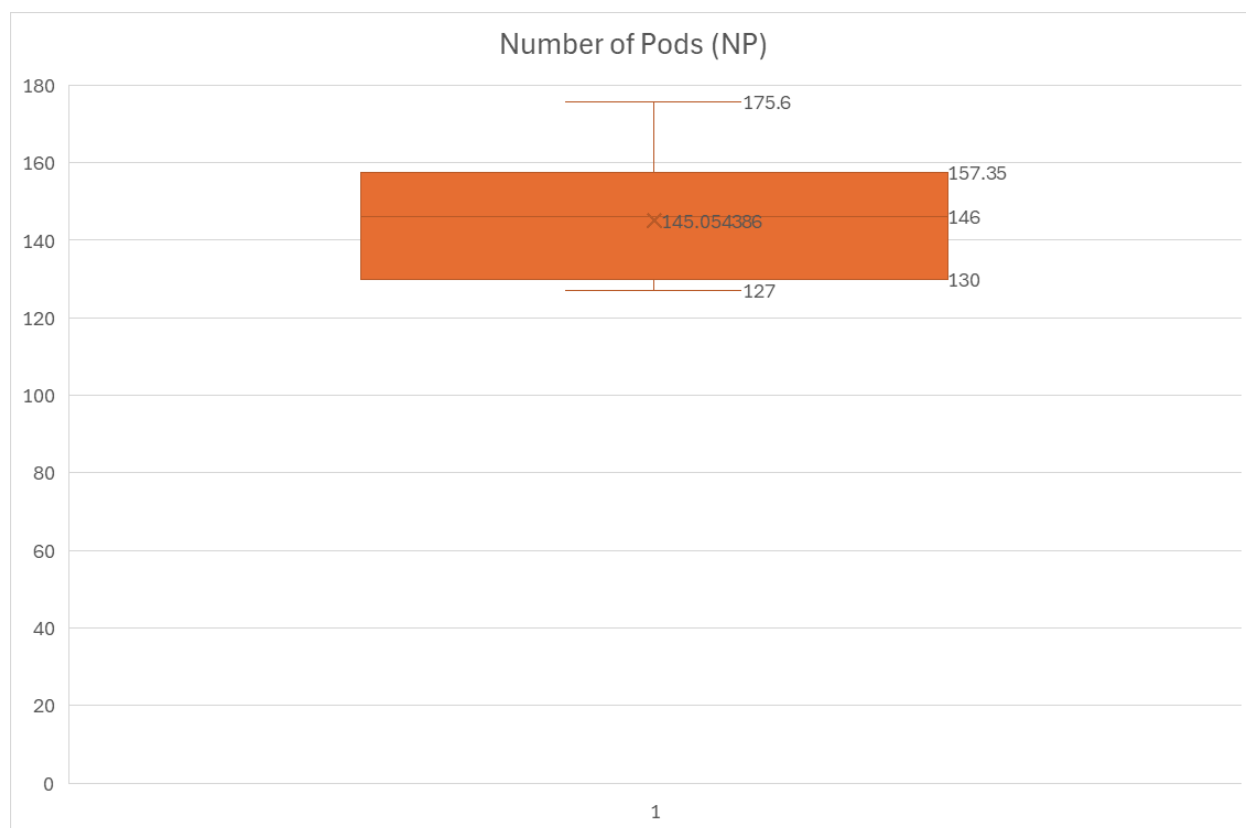


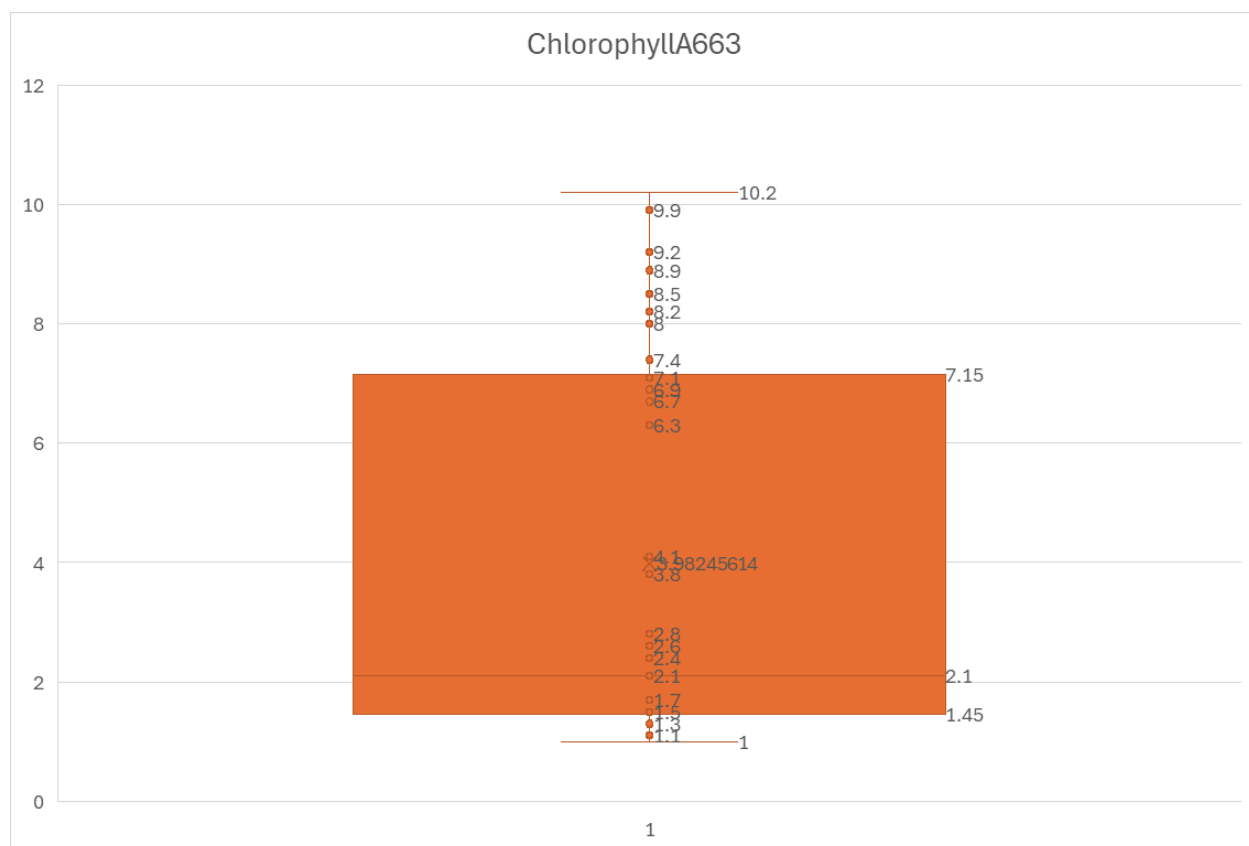
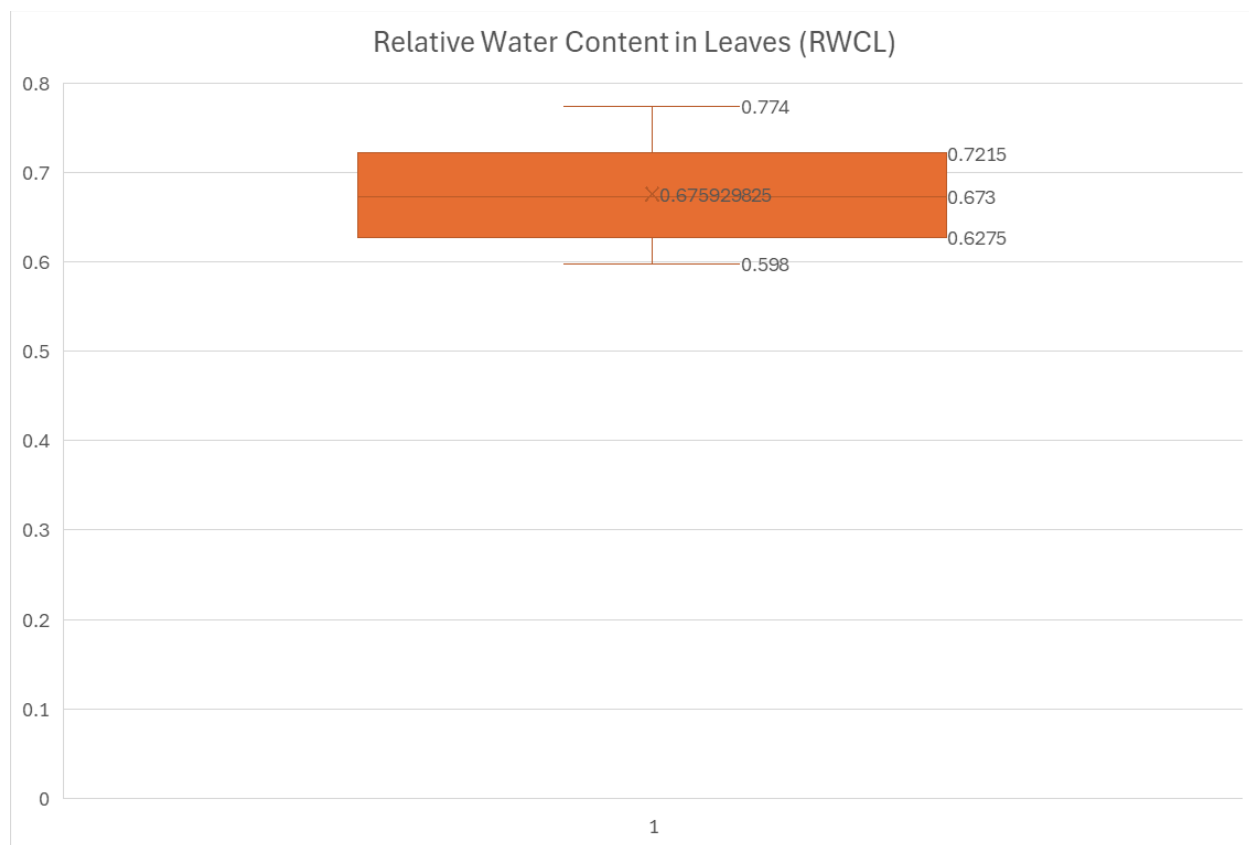
B2. Box Plots

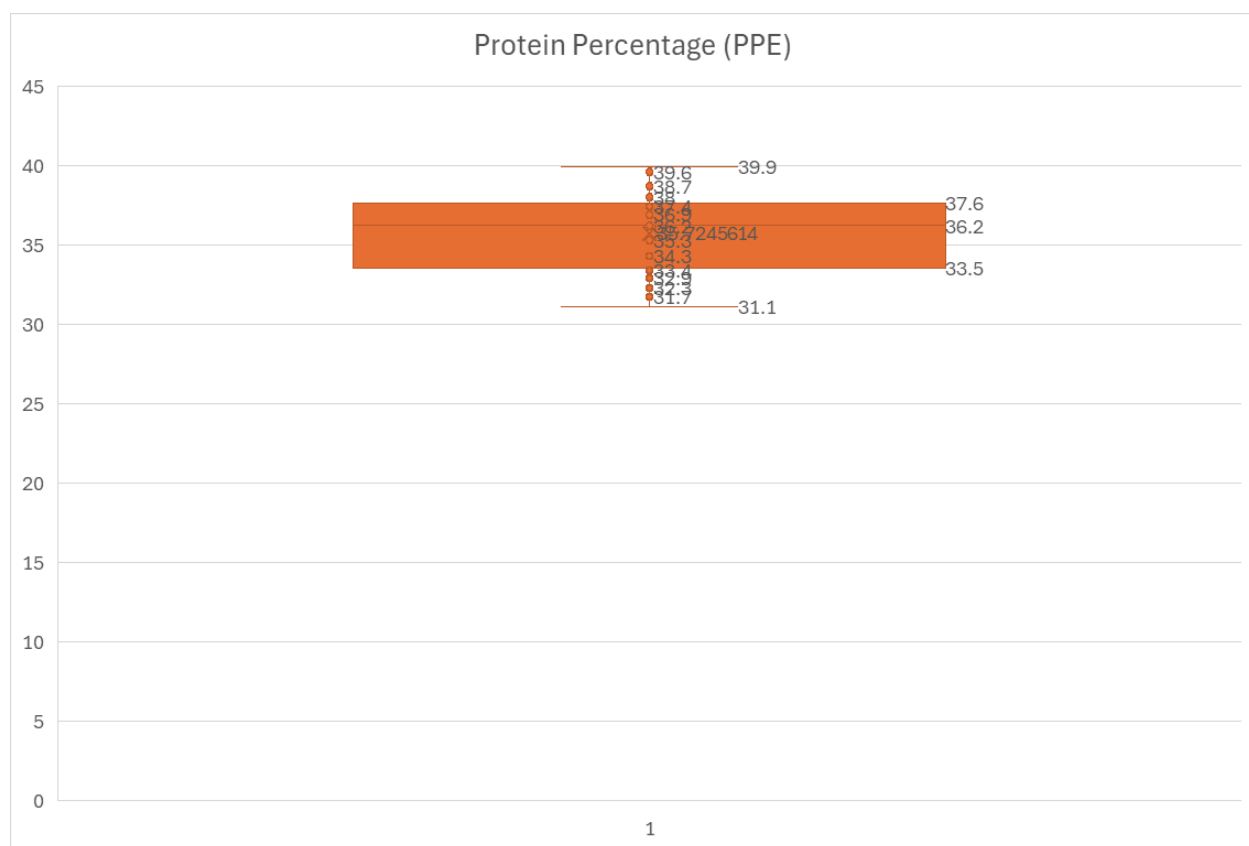
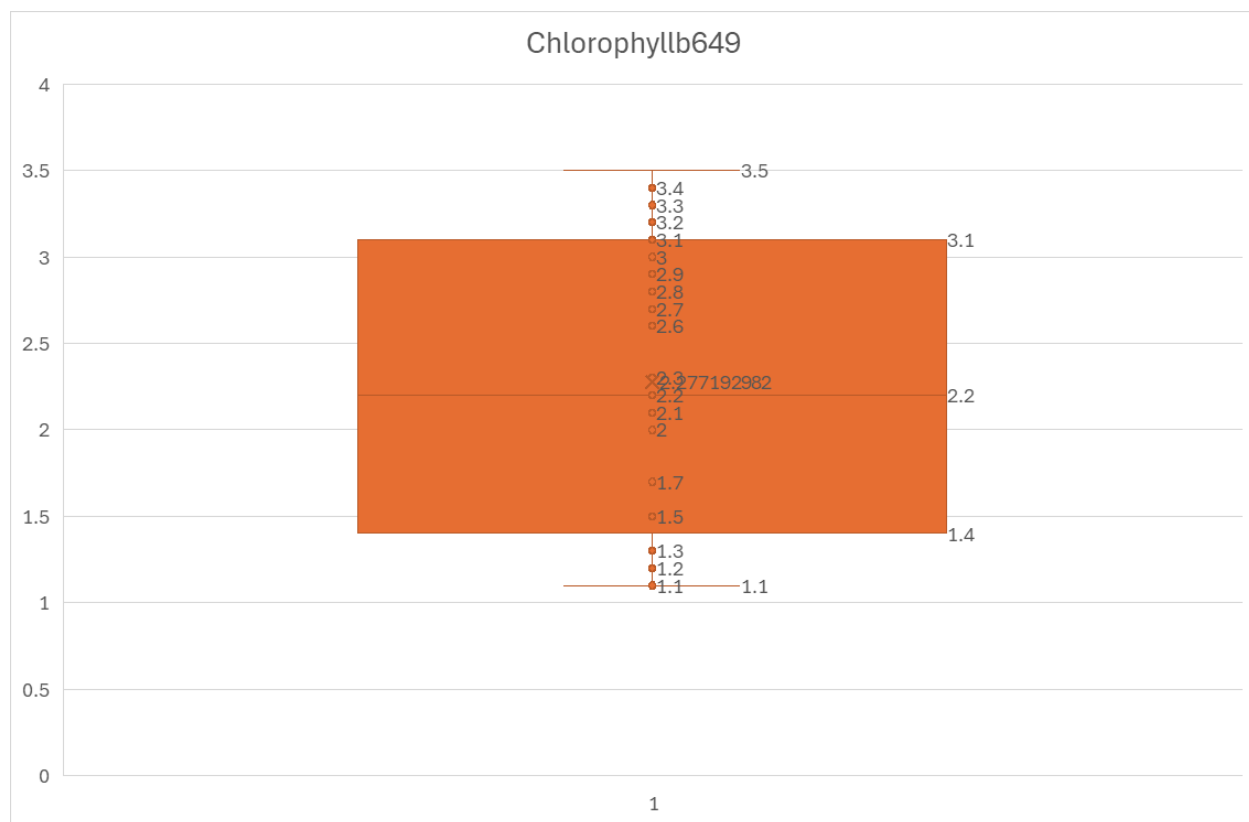


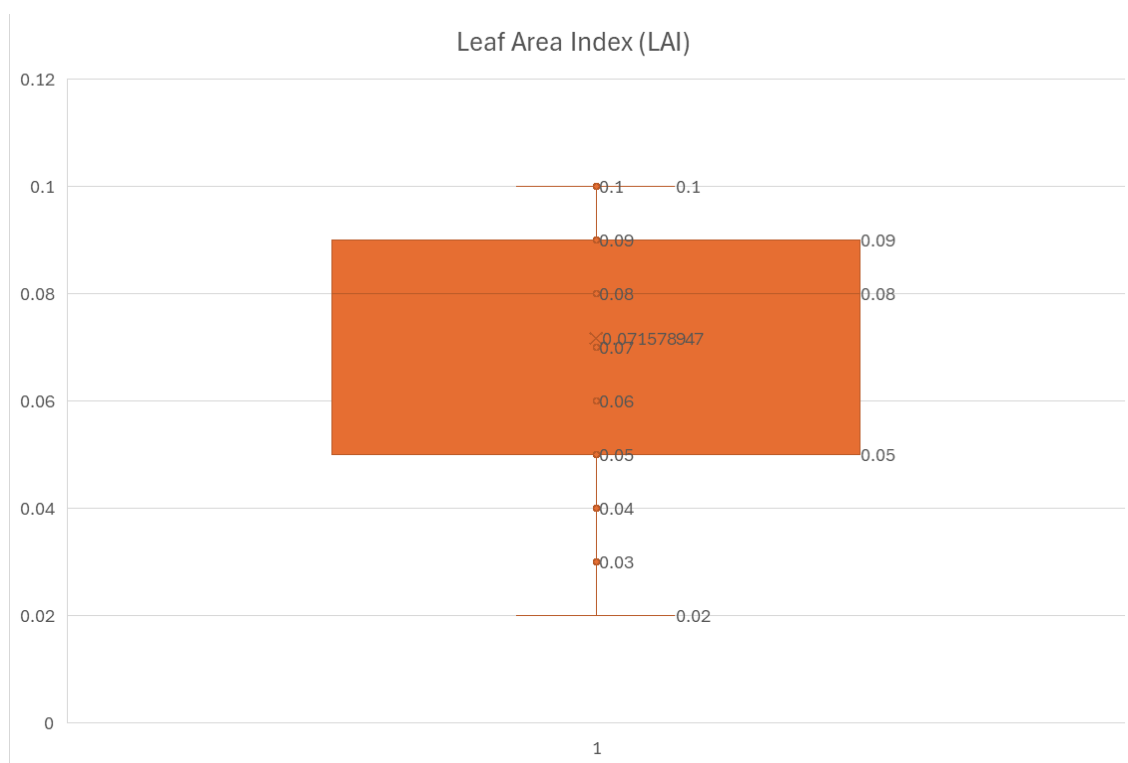
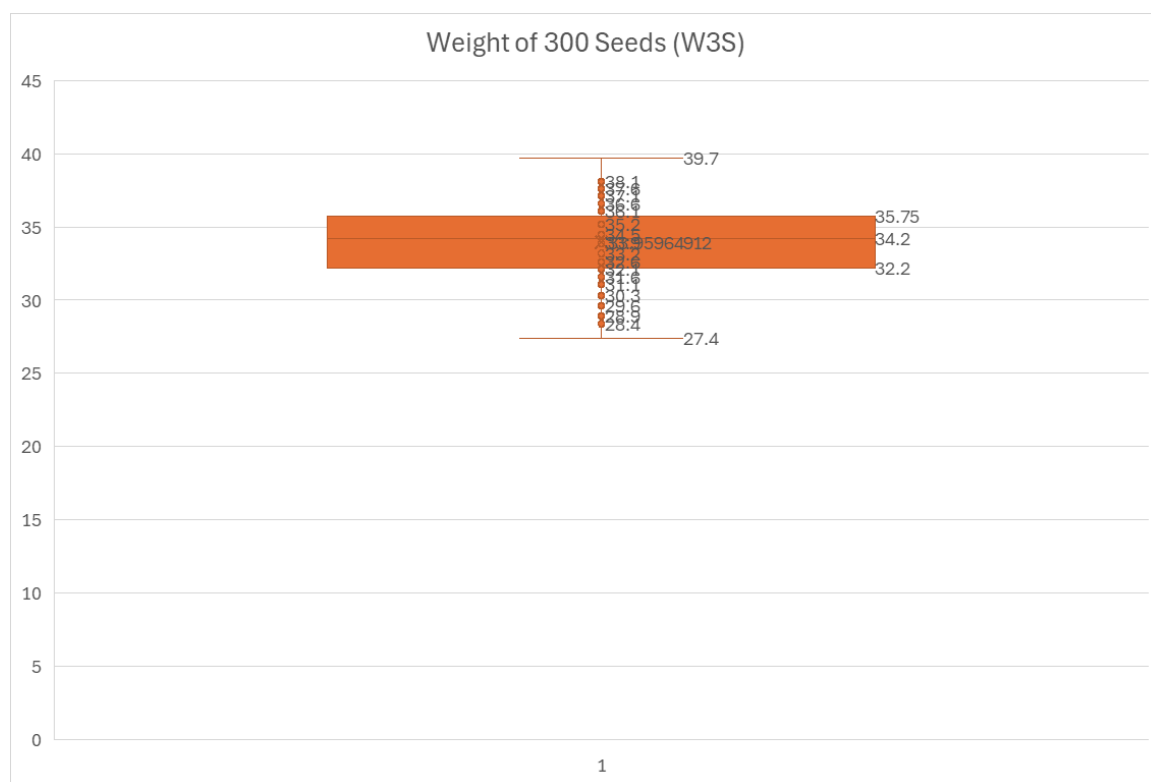
The BW box plot shows a median of 90 and a mean of 97.49. Most of its values can be analyzed by looking at its quartiles. BW values mostly span between 76 to 123.5. Its minimum value is 63, and the maximum value is 138.

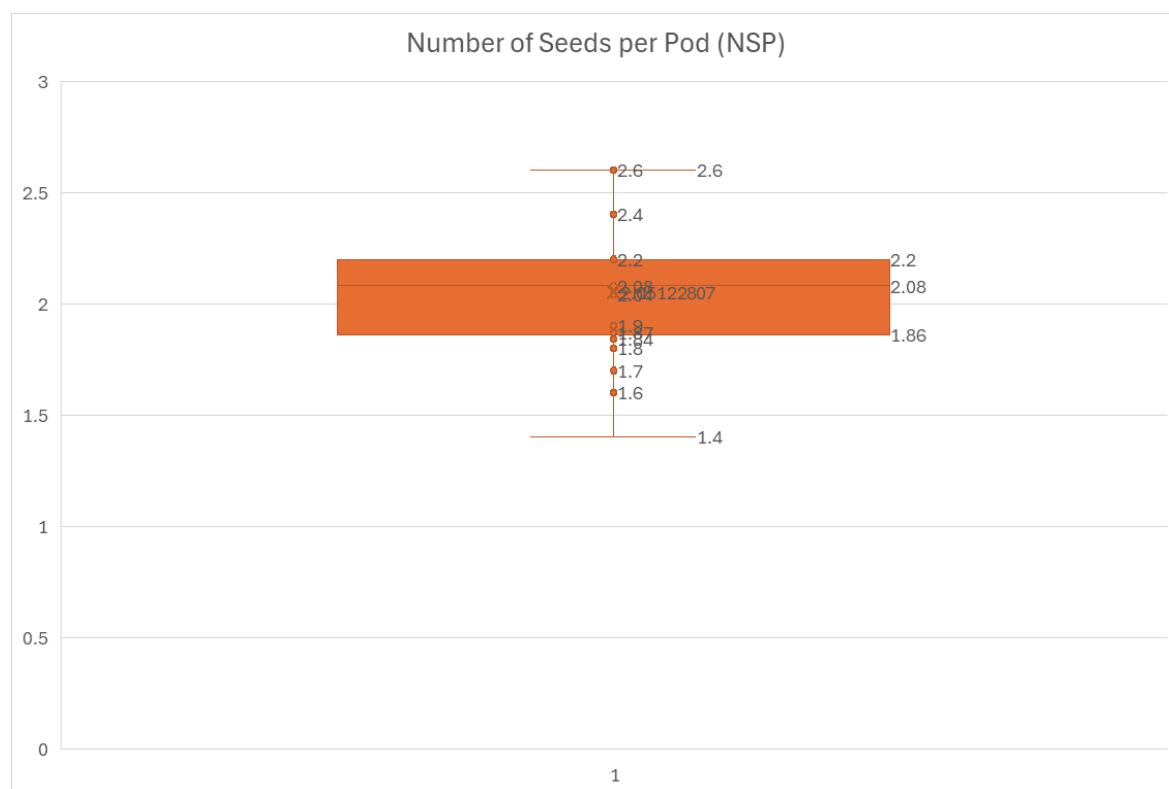
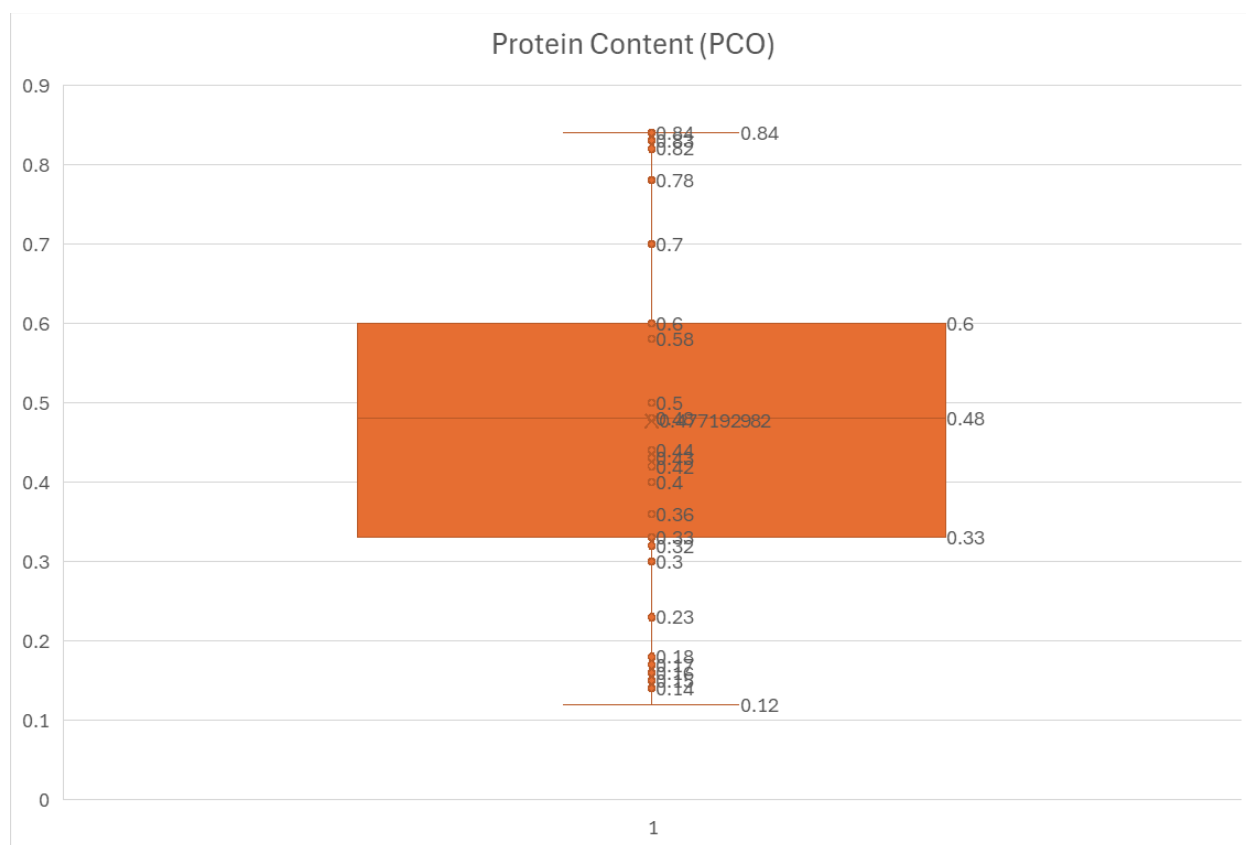






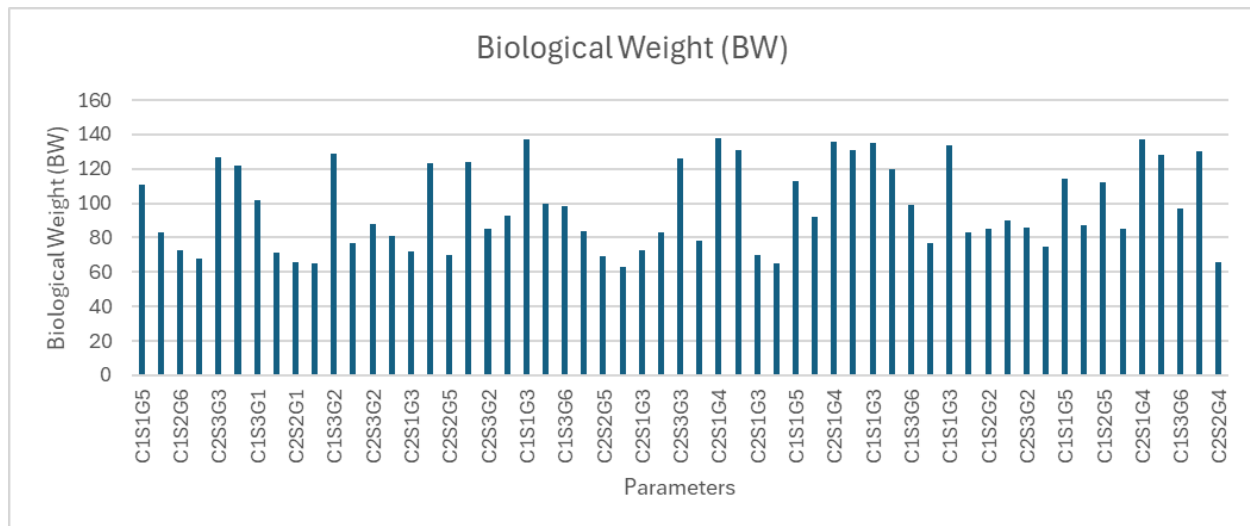




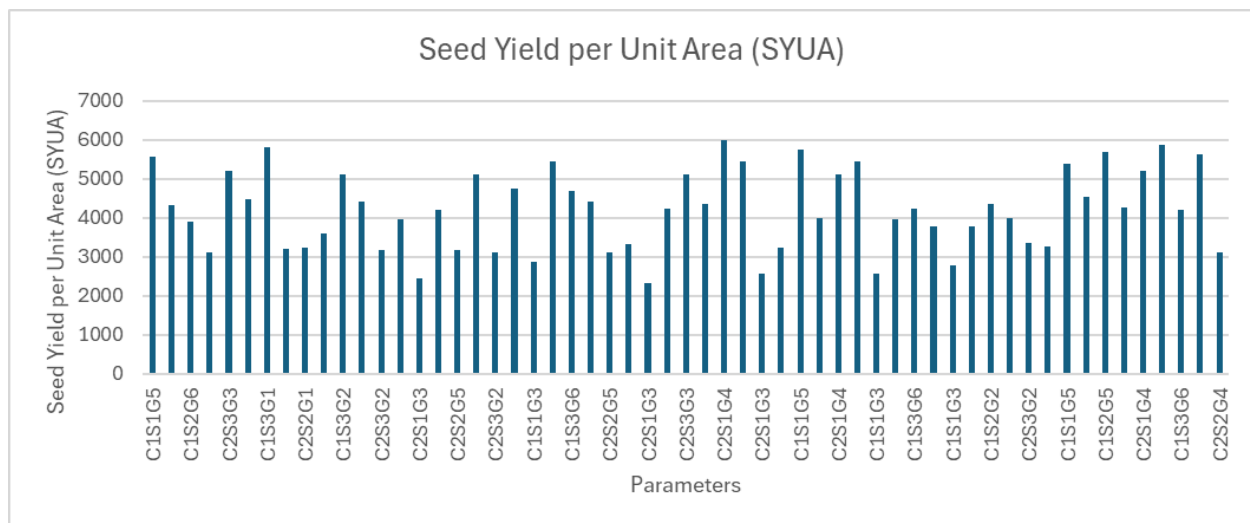


B3. Bar Charts

The x-axis used in the bar charts contains the parameters of the soybeans. These parameters have different combinations that distinguish one soybean from another, whether in terms of environmental location, treatments, etc. These bar charts help evaluate the differences in results with respect to their parameters, better identifying which ones produce the most desirable outcomes.

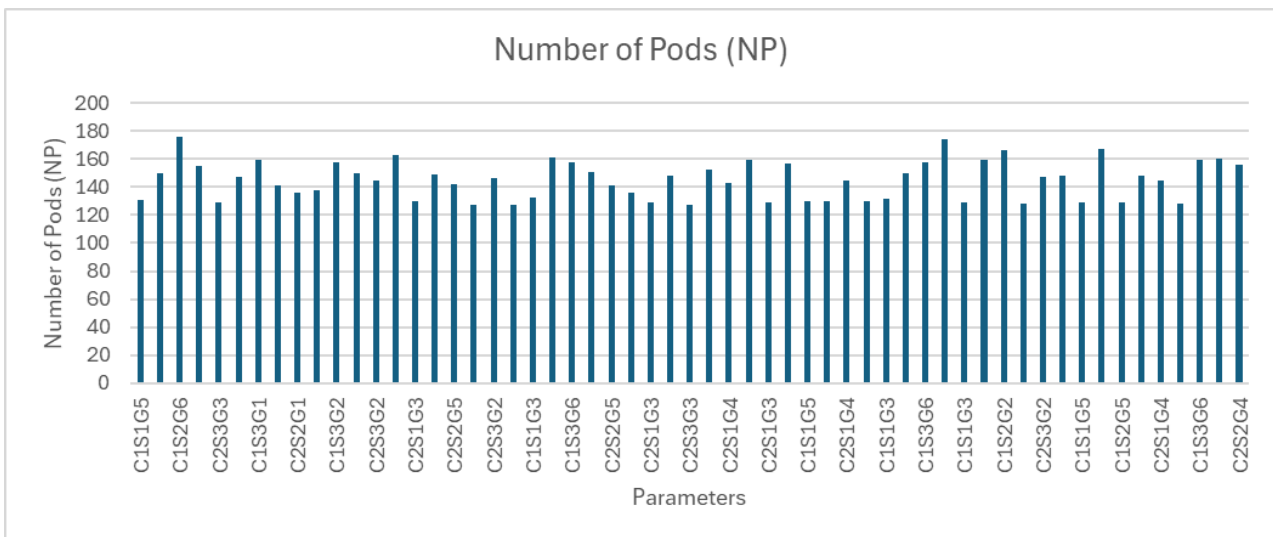
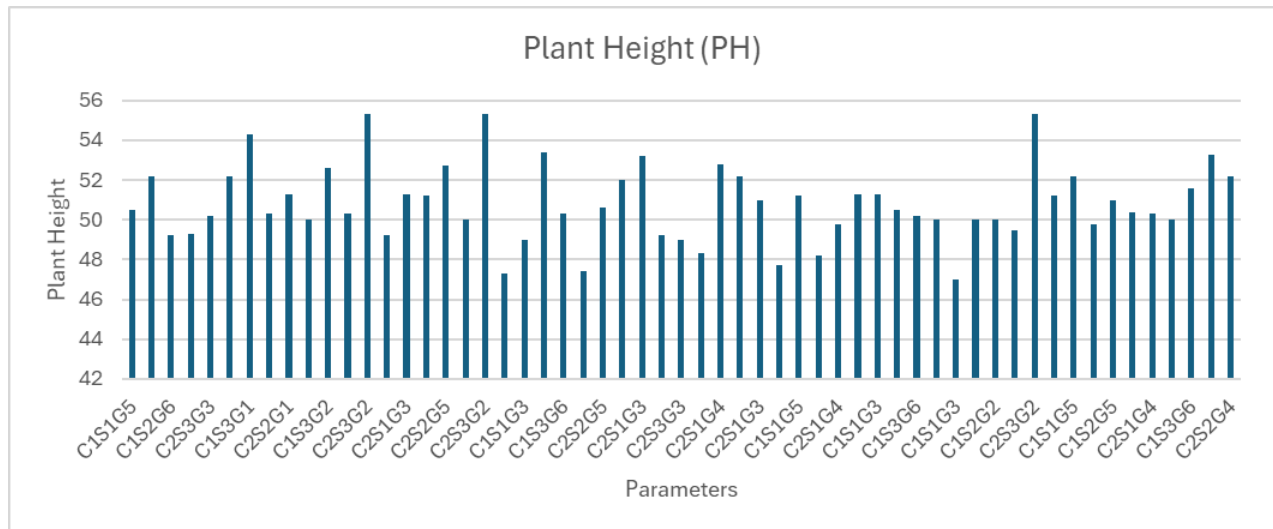


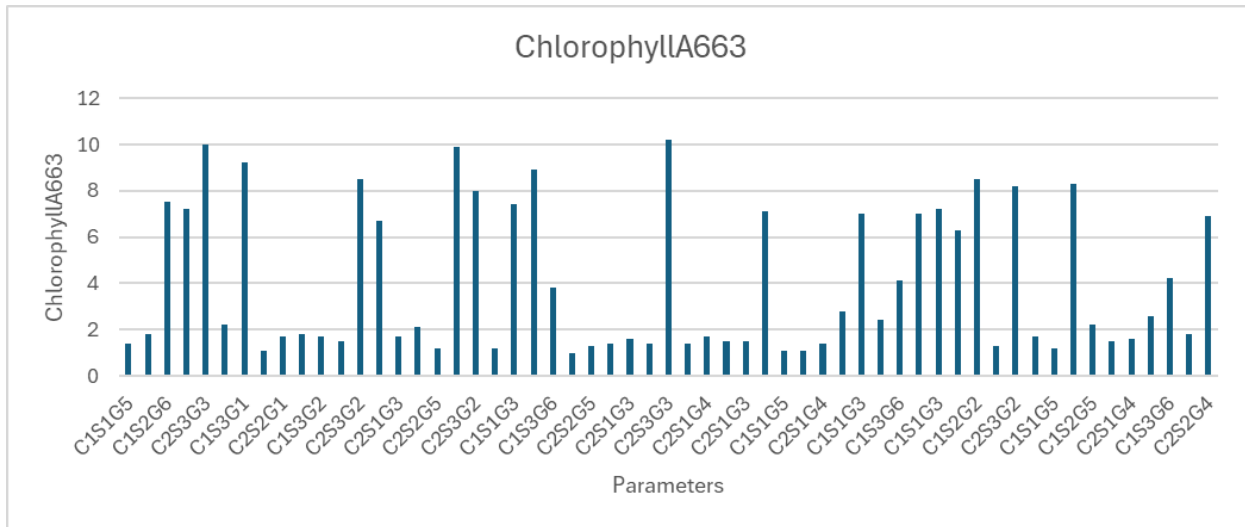
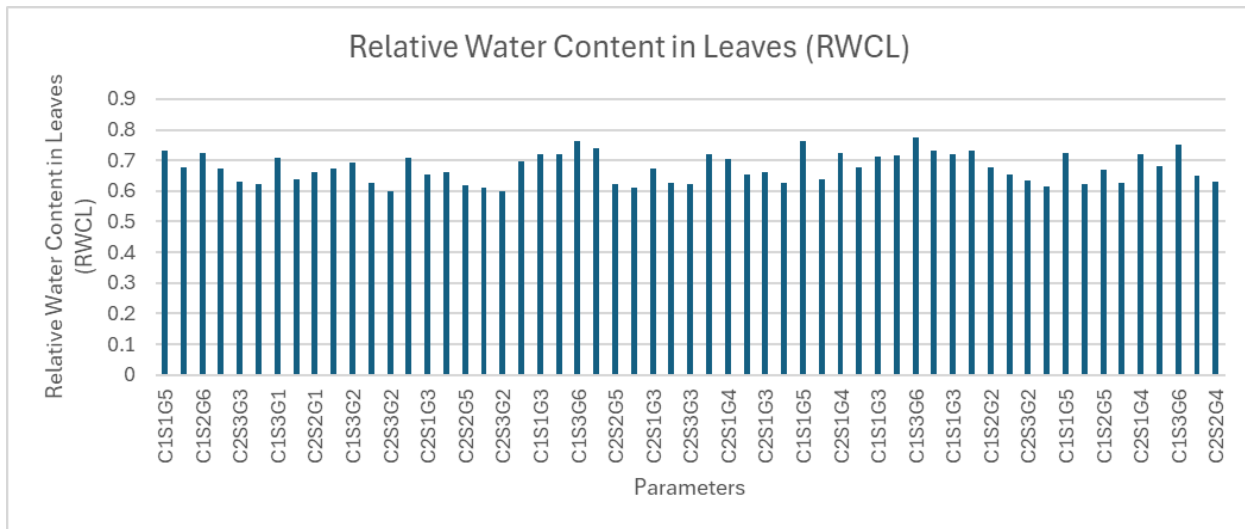
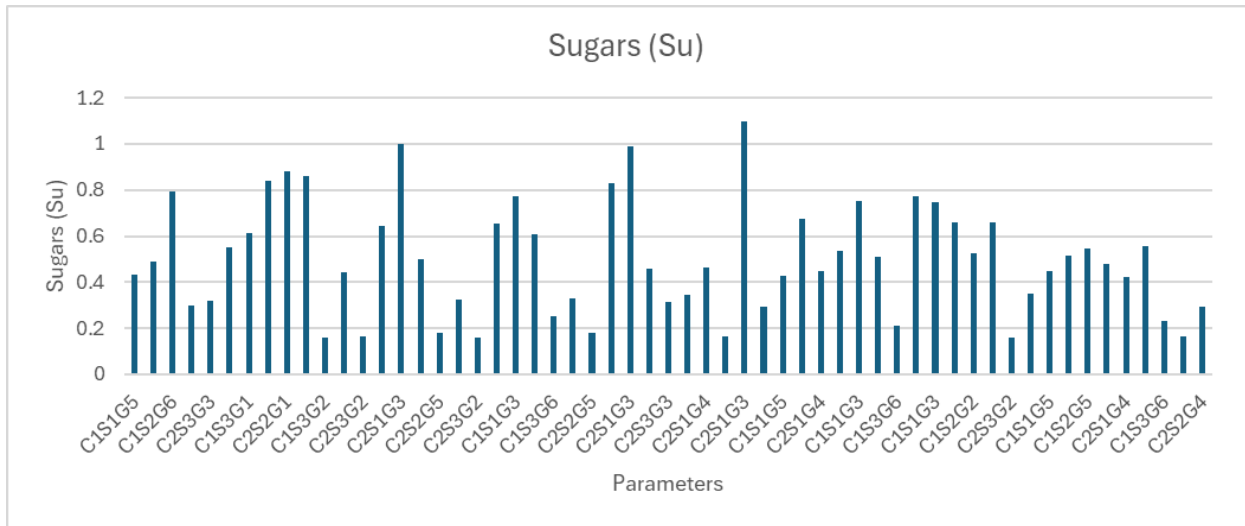
As seen in the BW bar chart, most of the values range from 60 to 130. Some of the parameters with the highest BW are C2S3G2 and C2S1G4.

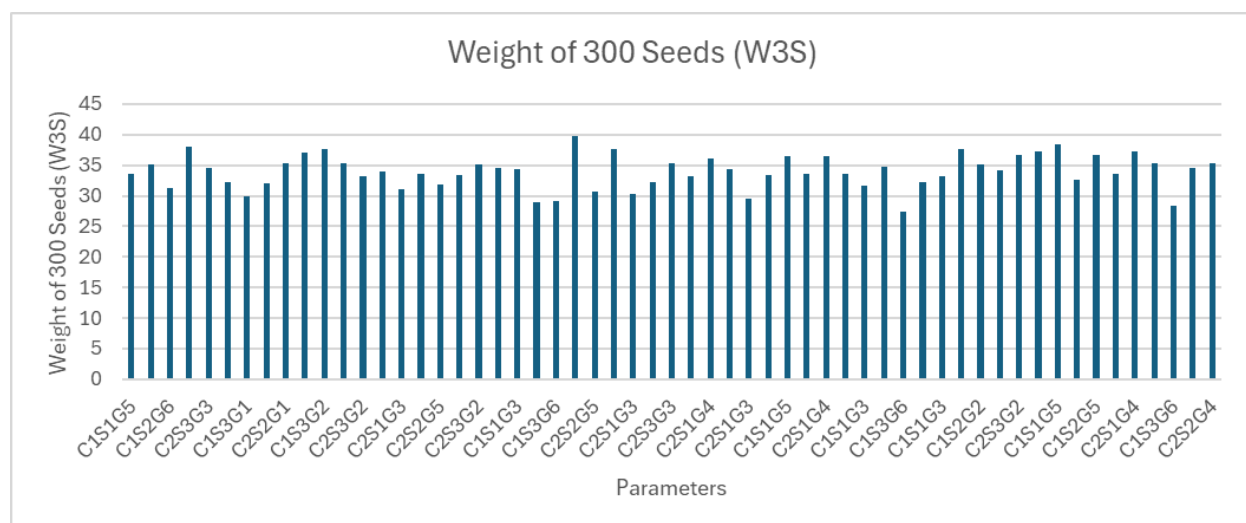
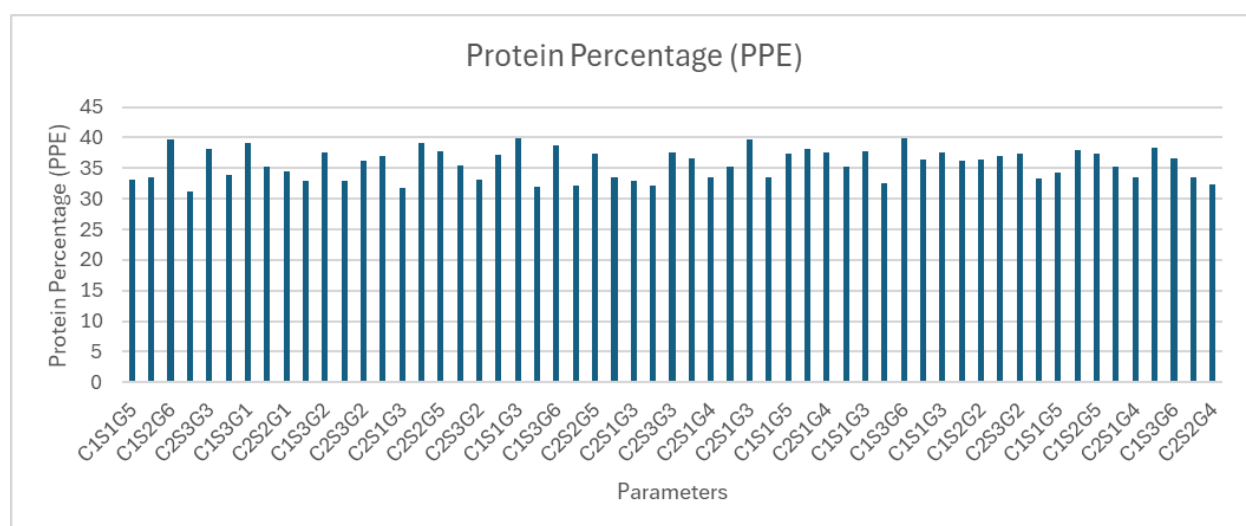
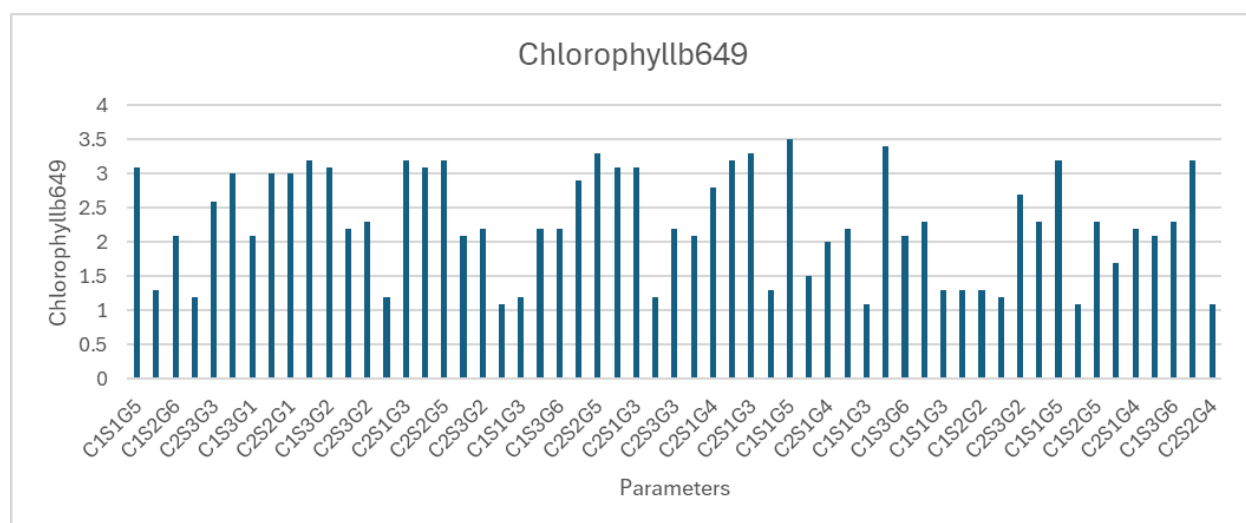


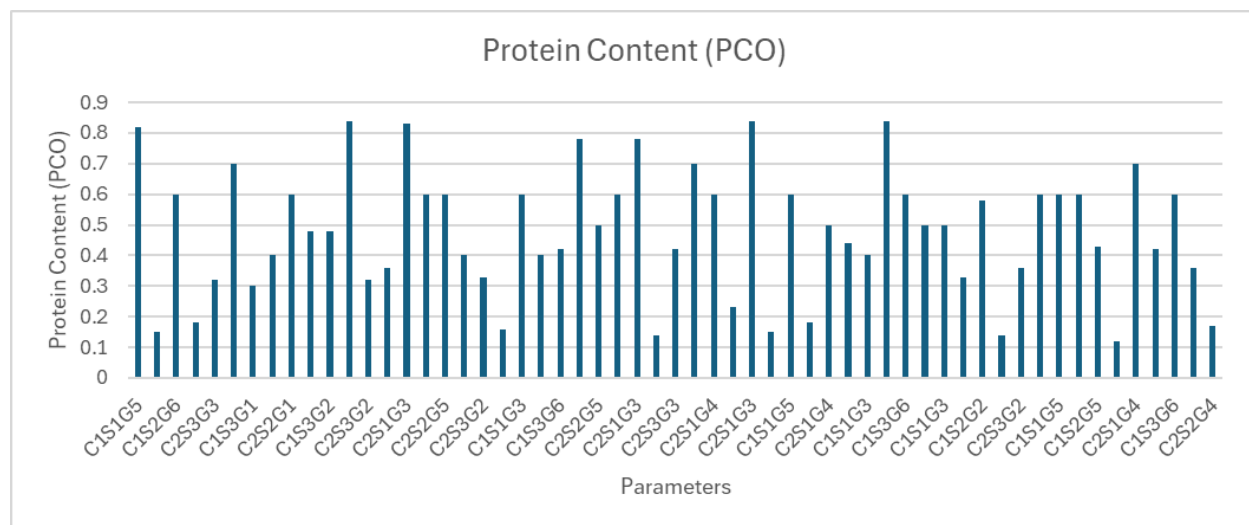
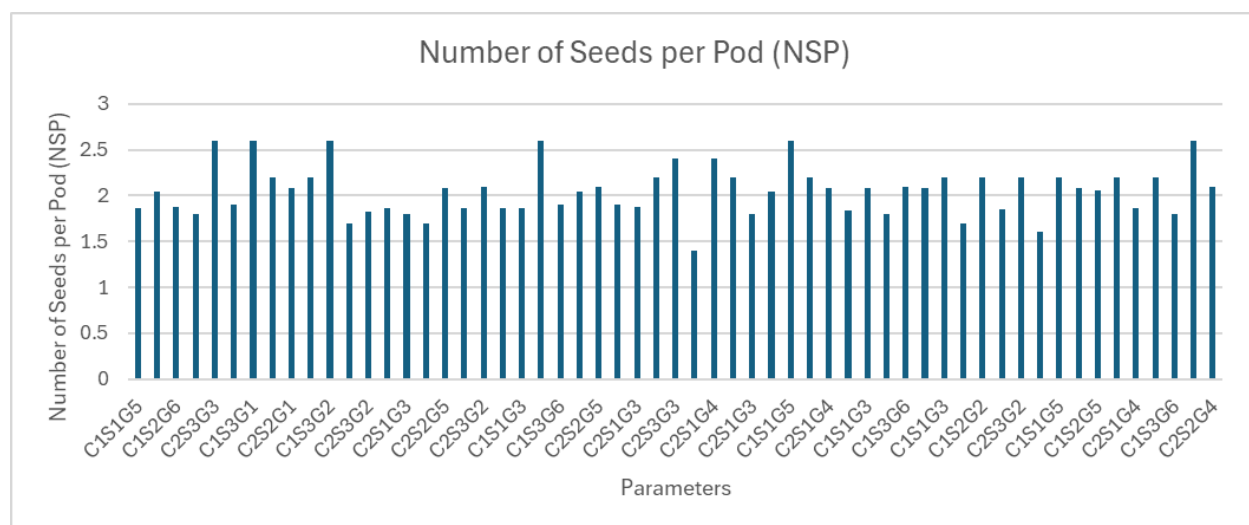
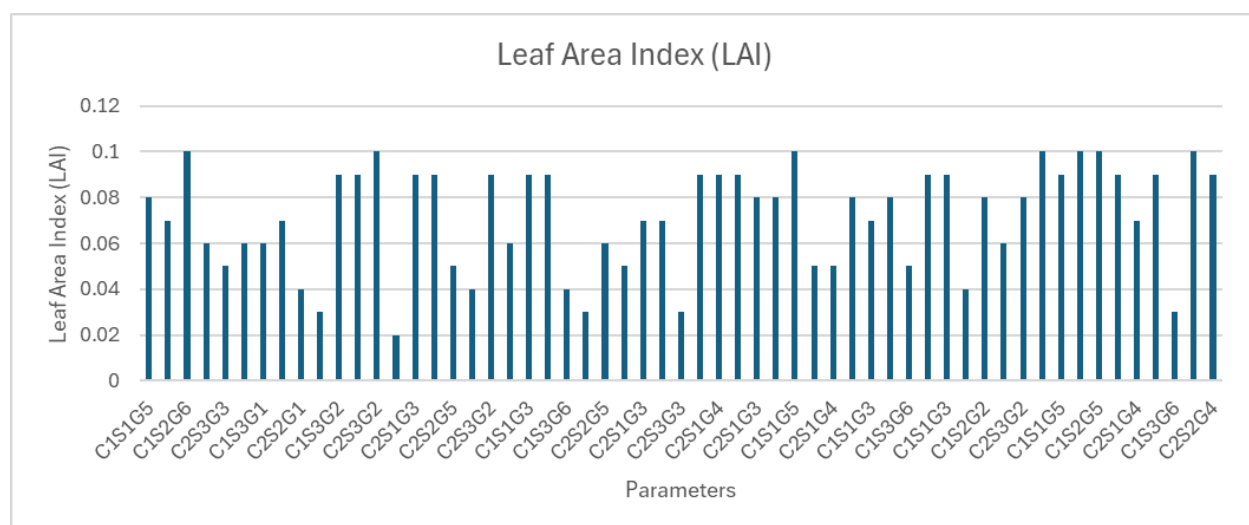
The SYUA bar chart shows that most seed yields are between 3000 and 6000. The highest yield includes C2S1G4, C1S1G4, and C2S3G2 with SYUAs between 5800 to 6000. Meanwhile, the lowest SYUA is 2500 to 3000.

Below are the bar charts of the rest of the factors considered in the dataset.



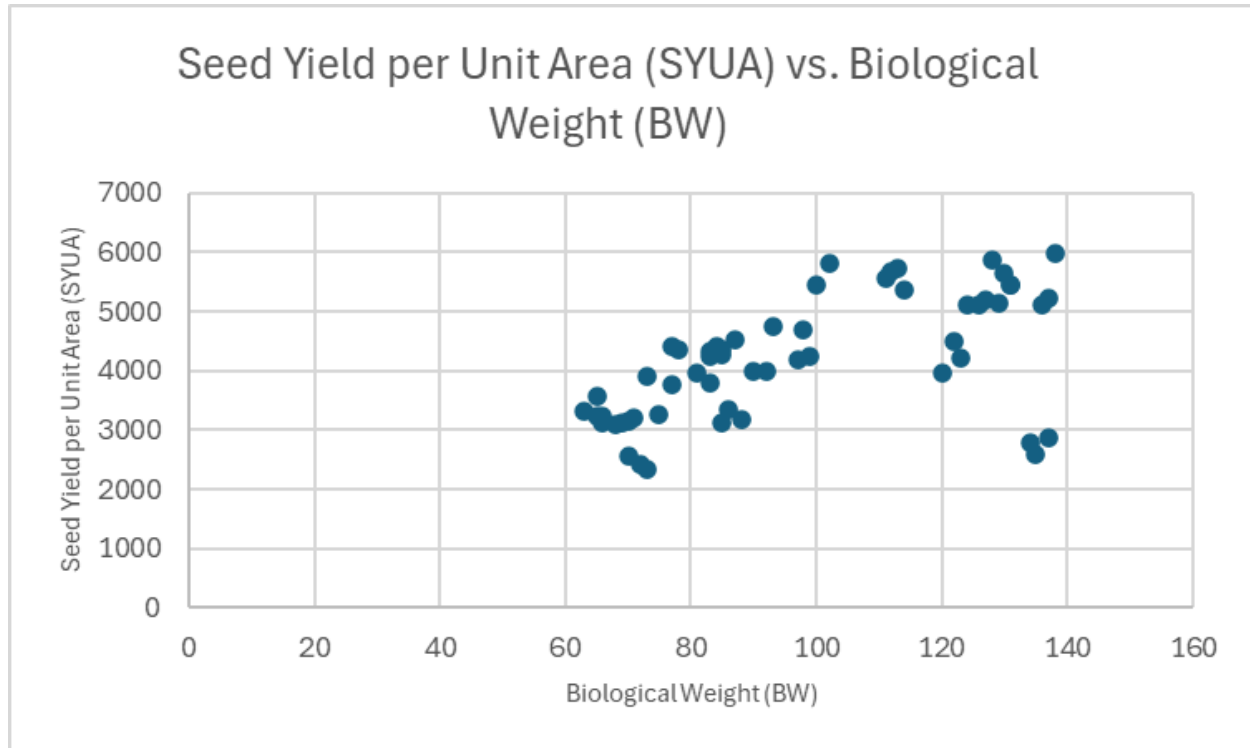






B4. Scatter Plots

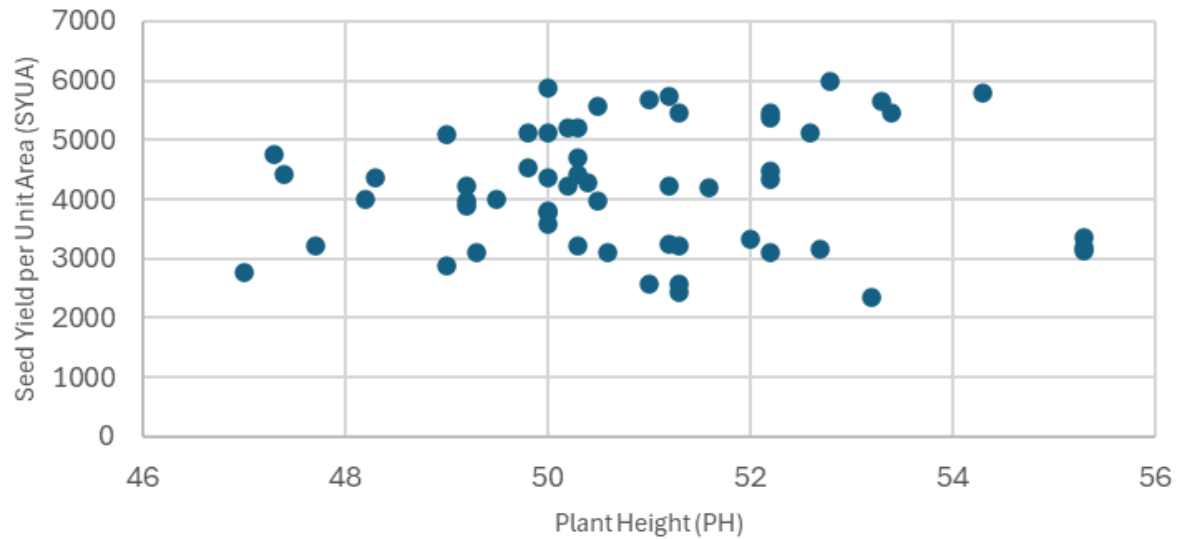
The scatter plots have a similar y-axis, which is the SYUA, because it is the dependent variable being focused on in this report. Therefore, using this can aid in better understanding the different factors affecting its results.



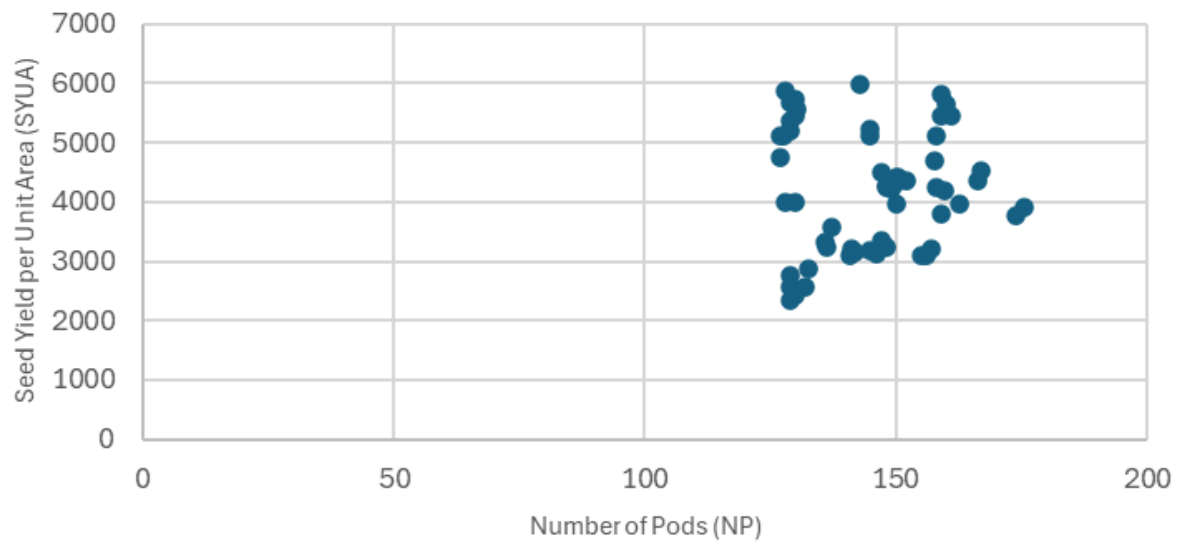
This scatter plot shows that there is a positive correlation between BW and SYUA. As BW increases, so does the SYUA. There is an upward trend with some variability.

Below are the bar charts of the rest of the factors considered in the dataset.

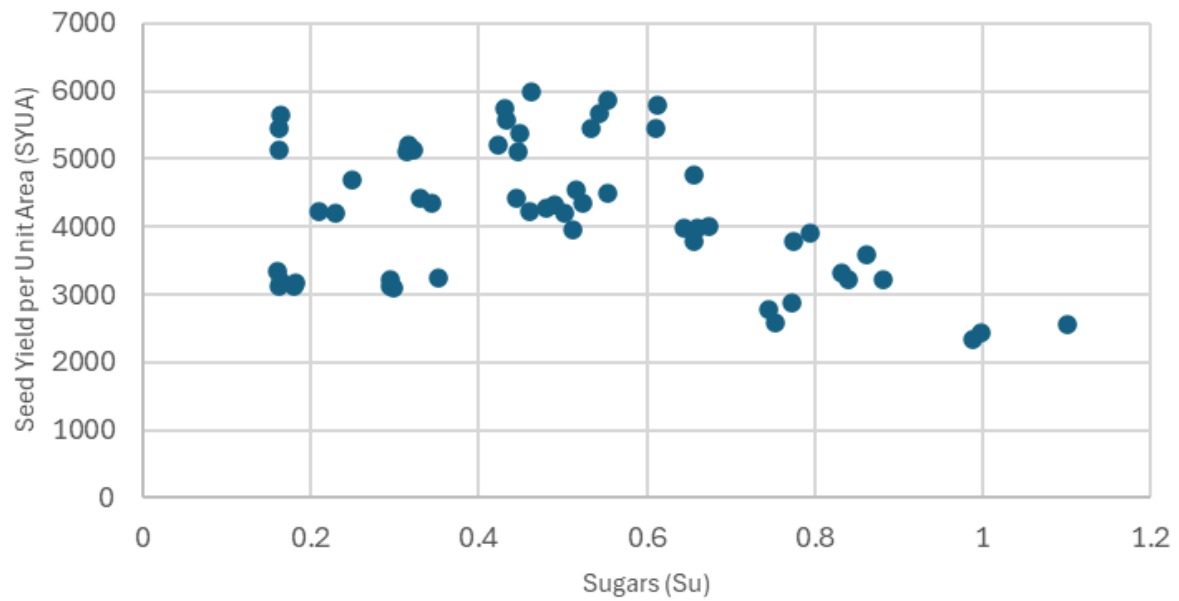
Seed Yield per Unit Area (SYUA) vs. Plant Height (PH)



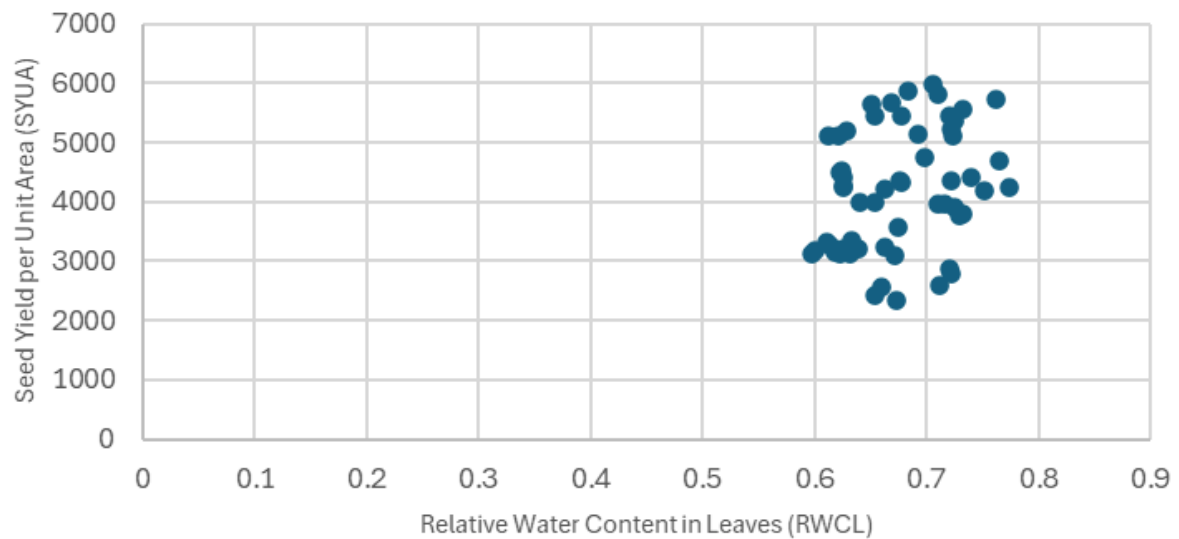
Seed Yield per Unit Area (SYUA) vs. Number of Pods (NP)



Seed Yield per Unit Area (SYUA) vs. Sugars (Su)



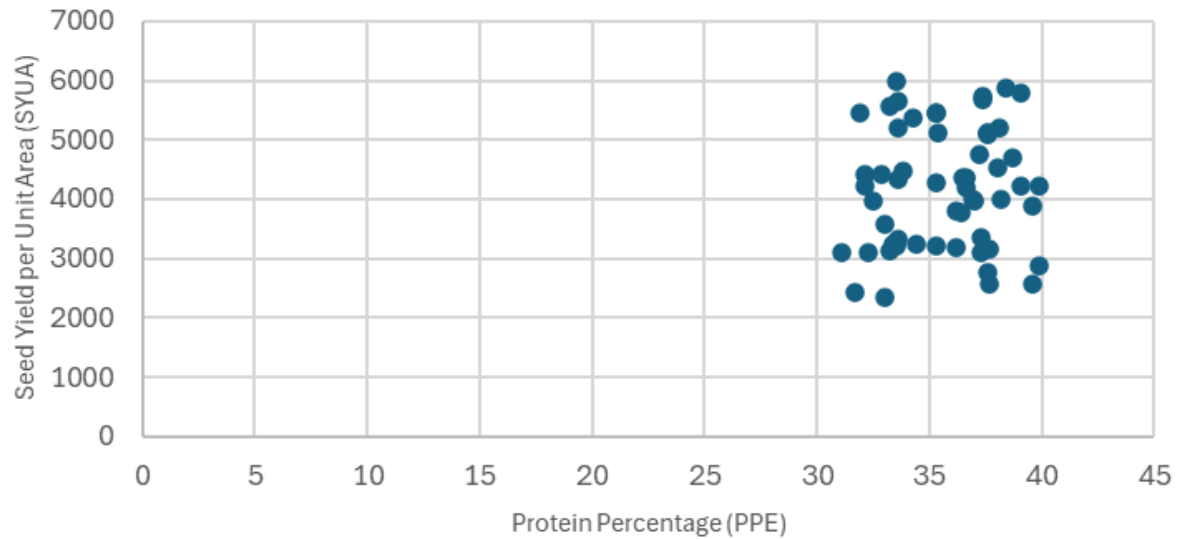
Seed Yield per Unit Area (SYUA) vs. Relative Water Content in Leaves (RWCL)



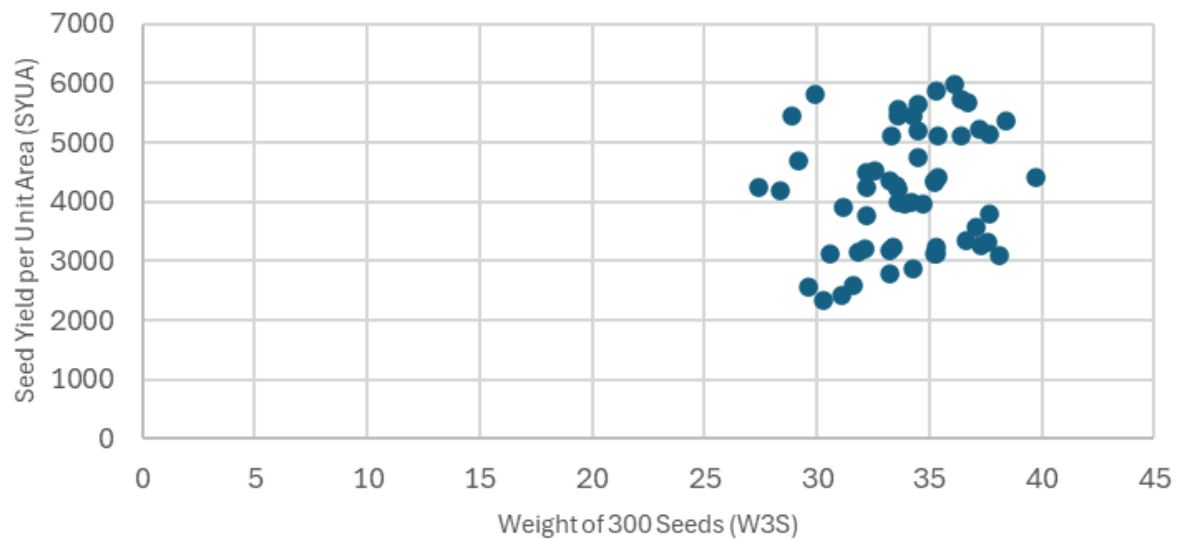
A scatter plot showing the relationship between Chlorophyll a content (ChlorophyllA663) on the x-axis and Seed Yield per Unit Area (SYUA) on the y-axis. The x-axis ranges from 0 to 12, and the y-axis ranges from 0 to 7000. The data points are clustered into two groups: one with low Chlorophyll a content (around 1.5) and high SYUA (around 4000-6000), and another with high Chlorophyll a content (around 7-10) and SYUA (around 3000-5000).

A scatter plot showing the relationship between Chlorophyllb649 (X-axis) and Seed Yield per Unit Area (SYUA) (Y-axis). The X-axis ranges from 0 to 4, and the Y-axis ranges from 0 to 7000. The data points are dark blue circles. The plot shows a positive correlation, with points generally increasing from left to right, though with some scatter.

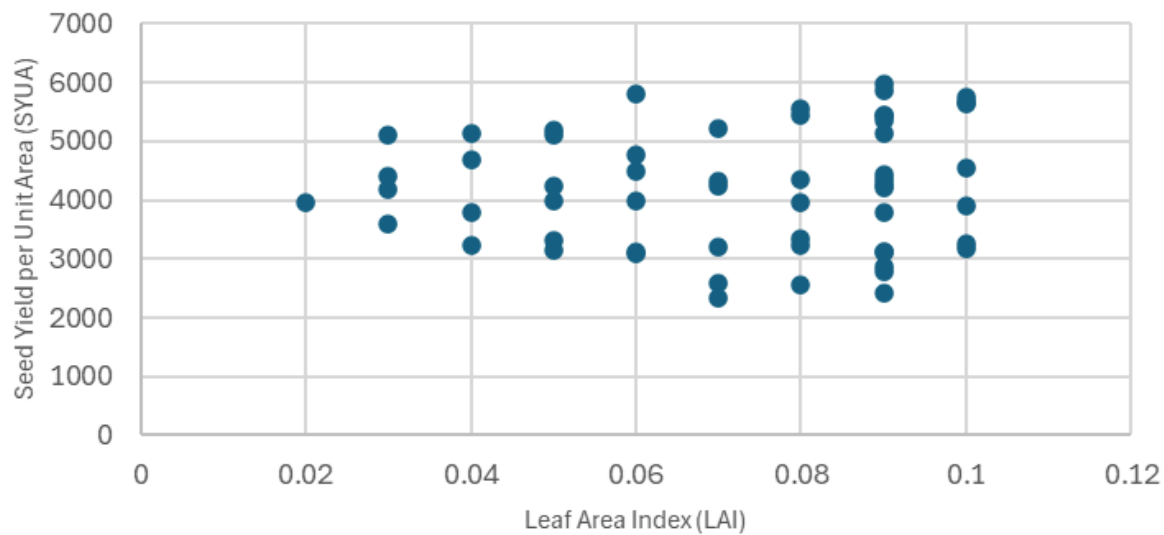
Seed Yield per Unit Area (SYUA) vs. Protein Percentage (PPE)



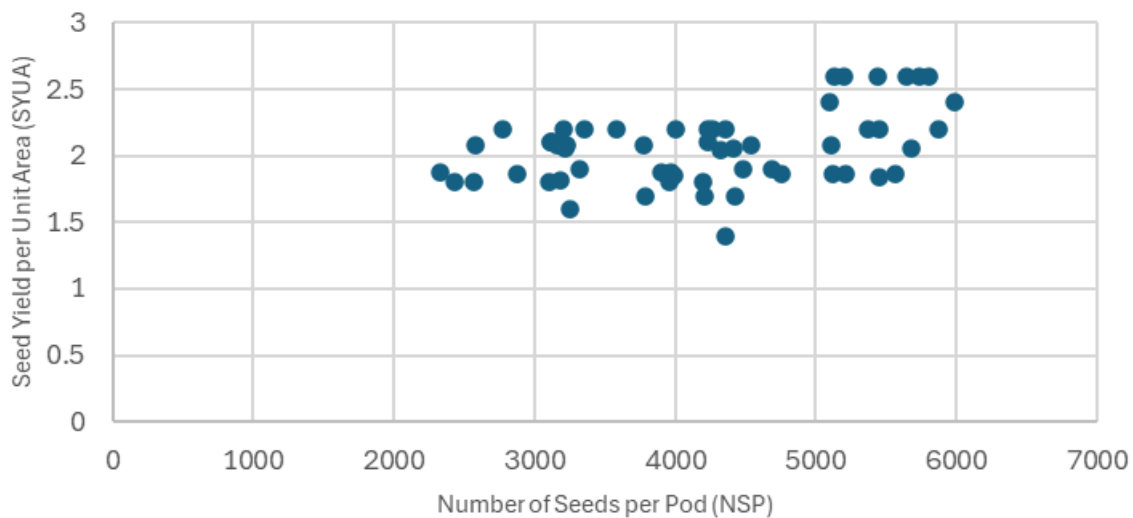
Seed Yield per Unit Area (SYUA) vs. Weight of 300 Seeds (W3S)

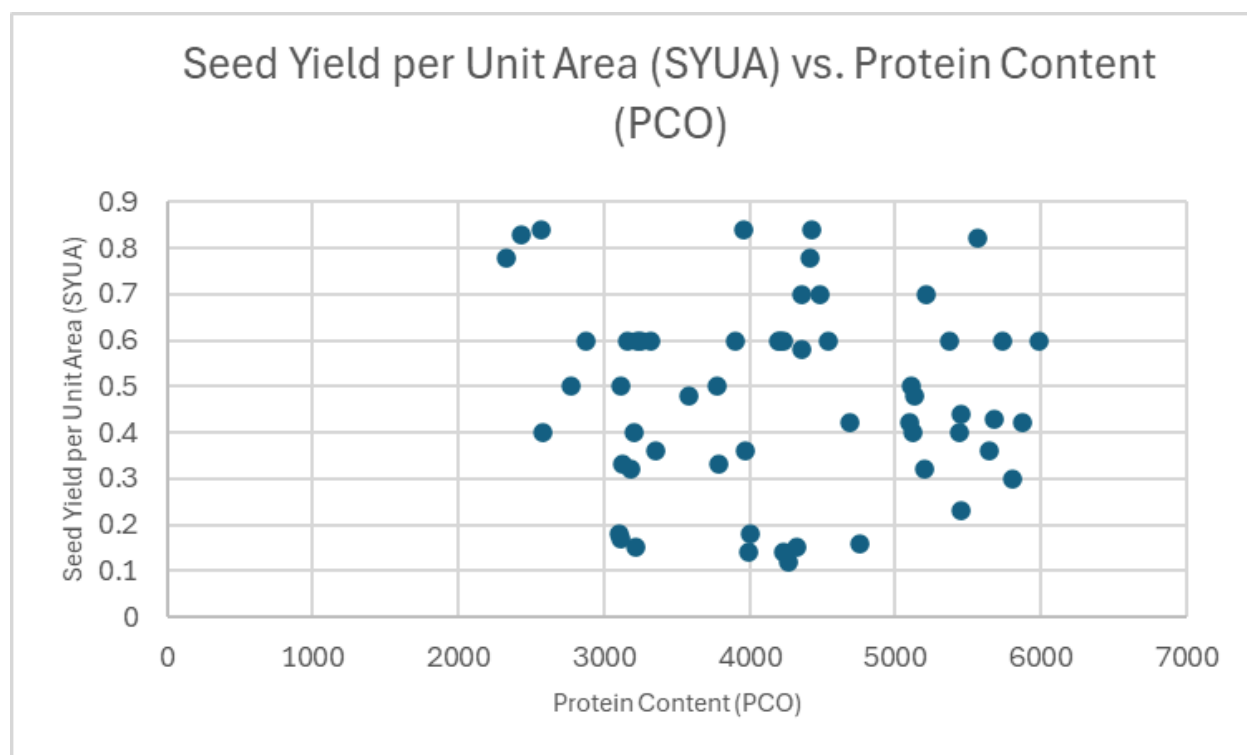


Seed Yield per Unit Area (SYUA) vs. Leaf Area Index (LAI)



Seed Yield per Unit Area (SYUA) vs. Number of Seeds per Pod (NSP)

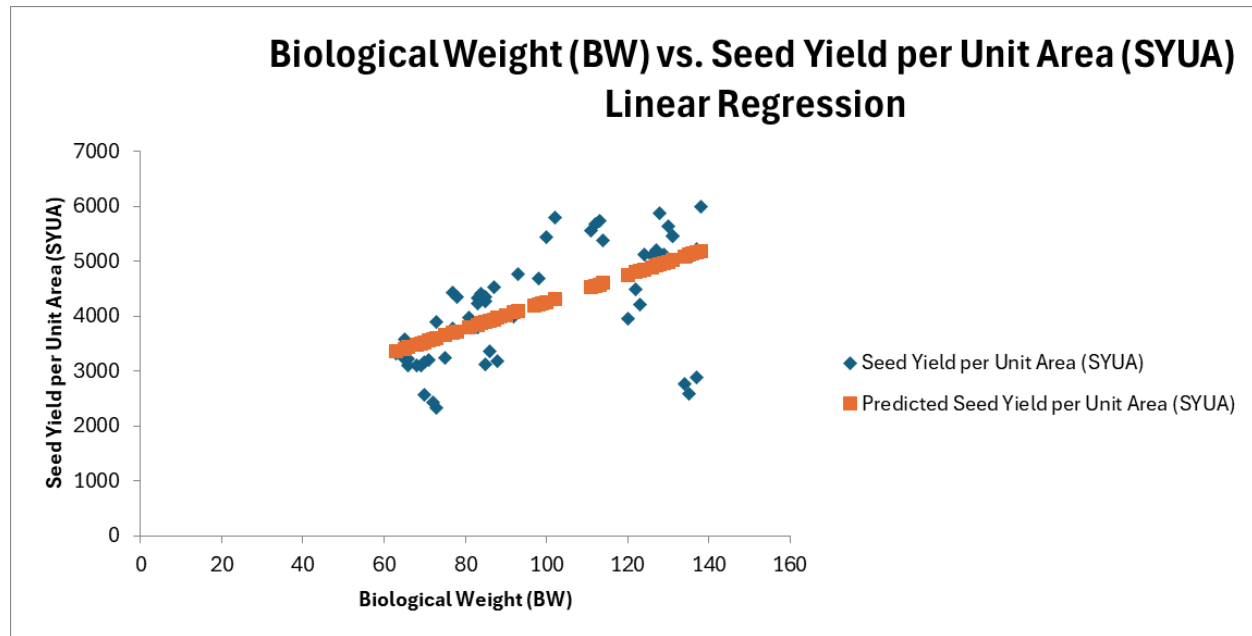




B5. Regression Analysis

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.59017							
R Square	0.348301							
Adjusted R	0.336451							
Standard E	830.8227							
Observations	57							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	20290193	20290193	29.39473	1.35E-06			
Residual	55	37964653	690266.4					
Total	56	58254846						
<i>Coefficients</i>		<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	1818.407	452.6203	4.01751	0.00018	911.3354	2725.478	911.3354	2725.478
Biological	24.41587	4.503369	5.42169	1.35E-06	15.39092	33.44083	15.39092	33.44083

The slope means that for a 1 increase in the Biological Weight (BW) of a soybean, the Seed Yield per Unit Area (SYUA) increases by 24.42. Meanwhile, the intercept means that when the BW is 0, which is an unlikely scenario, the SYUA is 1818.41.



In terms of correlation, $r = 0.59017$ means that the Biological Weight and Seed Yield per Unit Area have a strong positive correlation. The R^2 of 0.3483 means that about 34.83% of the variability in SYUA data is explainable by the BW, and the remaining 65.17% is explainable by other factors.

C. Results from Hypothesis Testing

To evaluate the statistical relationship between **Biological Weight (BW)** and **Seed Yield per Unit Area (SYUA)** in soybean crops, a **simple linear regression model** was employed. The goal was to determine whether BW can significantly predict SYUA, which would provide evidence that heavier soybean plants tend to produce more seeds in a given area.

Term	Coefficient	Standard Error	t-Statistic	p-Value	Decision ($\alpha=0.05$)
Intercept	1818.41	452.62	4.02	0.00018017	Fail to Reject H_0
Biological Weight (BW)	24.42	4.50	5.42	1.3547E-06	Reject H_0

SYUA has a significant positive effect on Biological Weight (BW) because the p-value = 1.35467E-06 is less than 0.05. Thus, we reject the null hypothesis for BW at

0.05 significance level. Moreover, the Intercept is also statistically significant ($p = 0.00018$) because it is less than 0.05.

Regression Model Summary:

- **Regression Equation:**
 $SYUA = 0.02 \times BW + 11.75$
- **Interpretation of Coefficients:**
 - **Slope ($\beta_1 = 0.02$):**
This indicates that for every additional unit increase in Biological Weight, the Seed Yield per Unit Area increases by approximately 0.02 units, assuming all other factors remain constant.
 - **Intercept ($\beta_0 = 11.75$):**
When Biological Weight is zero, the predicted baseline seed yield per unit area is 11.75. While this value has no practical agronomic meaning (as $BW = 0$ is unrealistic), it's essential for model calculation.
- **Coefficient of Determination ($R^2 = 0.488$):**
This tells us that **48.8% of the variability** in seed yield outcomes is **explained by variations in biological weight** alone. This is a moderately strong relationship in biological data, where multiple interacting variables often influence outcomes.
- **Pearson Correlation ($r \approx 0.6987$):**
The positive correlation confirms a **moderate-to-strong linear association** between BW and SYUA, validating the visual trend observed in scatter plots.

Hypothesis Test for Regression Coefficient:

- **Null Hypothesis (H_0):**
Biological Weight has no effect on Seed Yield per Unit Area ($\beta_1 = 0$)
- **Alternative Hypothesis (H_1):**
Biological Weight has a significant effect on Seed Yield per Unit Area ($\beta_1 \neq 0$)
- **Test Used:**
A **t-test** was conducted on the regression coefficient (β_1) to determine statistical significance.
- **p-value Result:**
The calculated **p-value was less than 0.05**, which is the standard threshold for a 95% confidence level.

D. Conclusions and Recommendations

Decision and Conclusion:

Because the **p-value < 0.05**, we **reject the null hypothesis (H_0)** and accept the alternative. This confirms that **Biological Weight has a statistically significant effect on Seed Yield per Unit Area**. In practical terms, this means efforts to increase BW through better farming techniques, genetic selection, or environmental optimization are likely to lead to measurable improvements in yield.

Implications:

BW can serve as a **predictive indicator** for seed productivity early in the growing season. While BW explains nearly half of the variation in SYUA, further studies should incorporate other factors (i.e. soil composition, rainfall patterns, pest presence) for a more holistic model. Based on the findings, the following strategies are recommended to maximize seed yield:

1. Enhance Plant Health through Agronomic Management:

- Improve irrigation scheduling to avoid both drought stress and overwatering.
- Use targeted fertilization based on soil nutrient profiles to ensure balanced nutrition.
- Conduct regular soil testing to maintain optimal pH and structure, thereby supporting stronger root systems and higher BW.

2. Breed or Select High-BW Varieties:

- Encourage agricultural research institutions and seed producers to develop or promote soybean cultivars with naturally higher biological weight.
- Use BW as a **selection criterion** in breeding programs aimed at improving seed yield potential.
- Track BW throughout the growing cycle using non-destructive measurement tools (e.g., drones or multispectral imagery).

3. Implement Precision Agriculture Technologies:

- Utilize **soil sensors, weather-based modeling, and remote sensing (e.g., drones or satellites)** to monitor real-time plant conditions.
- Integrate data platforms that allow farmers to act on BW-related trends by adjusting inputs dynamically throughout the season.
- Use machine learning models trained on datasets like this to make real-time yield predictions and input recommendations.

4. Expand Multivariable Analysis in Future Research:

- While BW is a key factor, future studies should include additional variables such as:
 - **Soil type and nutrient density**

- **Microclimatic variations (humidity, wind, temperature extremes)**

- **Disease resistance and pest pressure**

- A multivariate regression or machine learning approach would help **refine predictive power** and tailor recommendations more accurately.

5. Educate Farmers and Field Technicians:

- Disseminate these findings through workshops, field demonstrations, and extension services.
- Train farm staff on how to recognize signs of poor biomass development early in the season and how to respond with corrective actions.

This comprehensive approach bridges data analysis with field application, helping stakeholders make **informed, yield-boosting decisions** across all phases of the soybean production cycle.